

Vectors and 3D Geometry

CHAPTER 1

Introduction to 3D Geometry 1.1–1.24

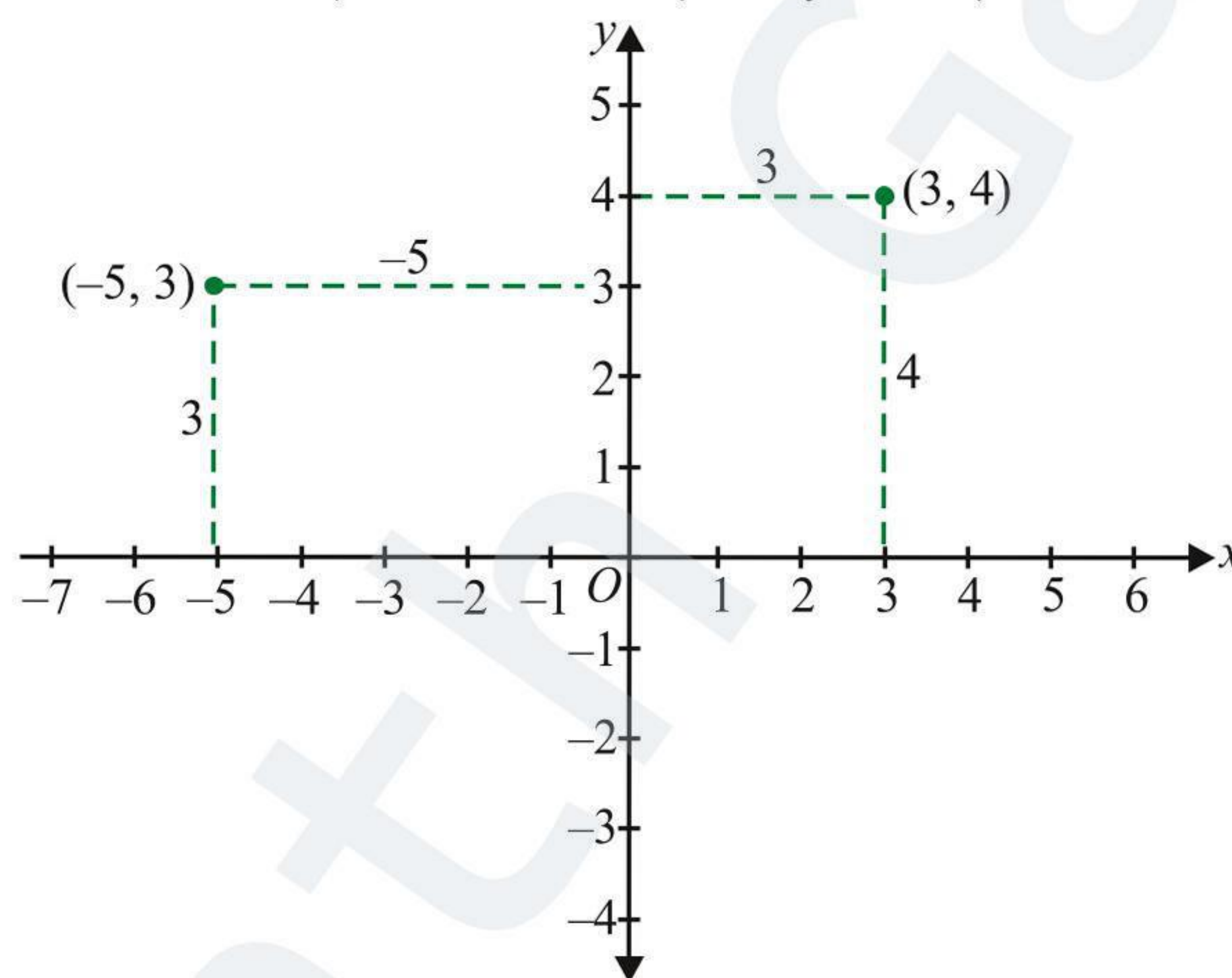
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1

Introduction to 3D Geometry

INTRODUCTION TO 3D COORDINATE SYSTEM

Let us recall the two-dimensional coordinate system. Consider point $P(x, y)$ on the cartesian plane which is formed by two perpendicular axes, x -axis (horizontal axis) and y -axis (vertical axis).



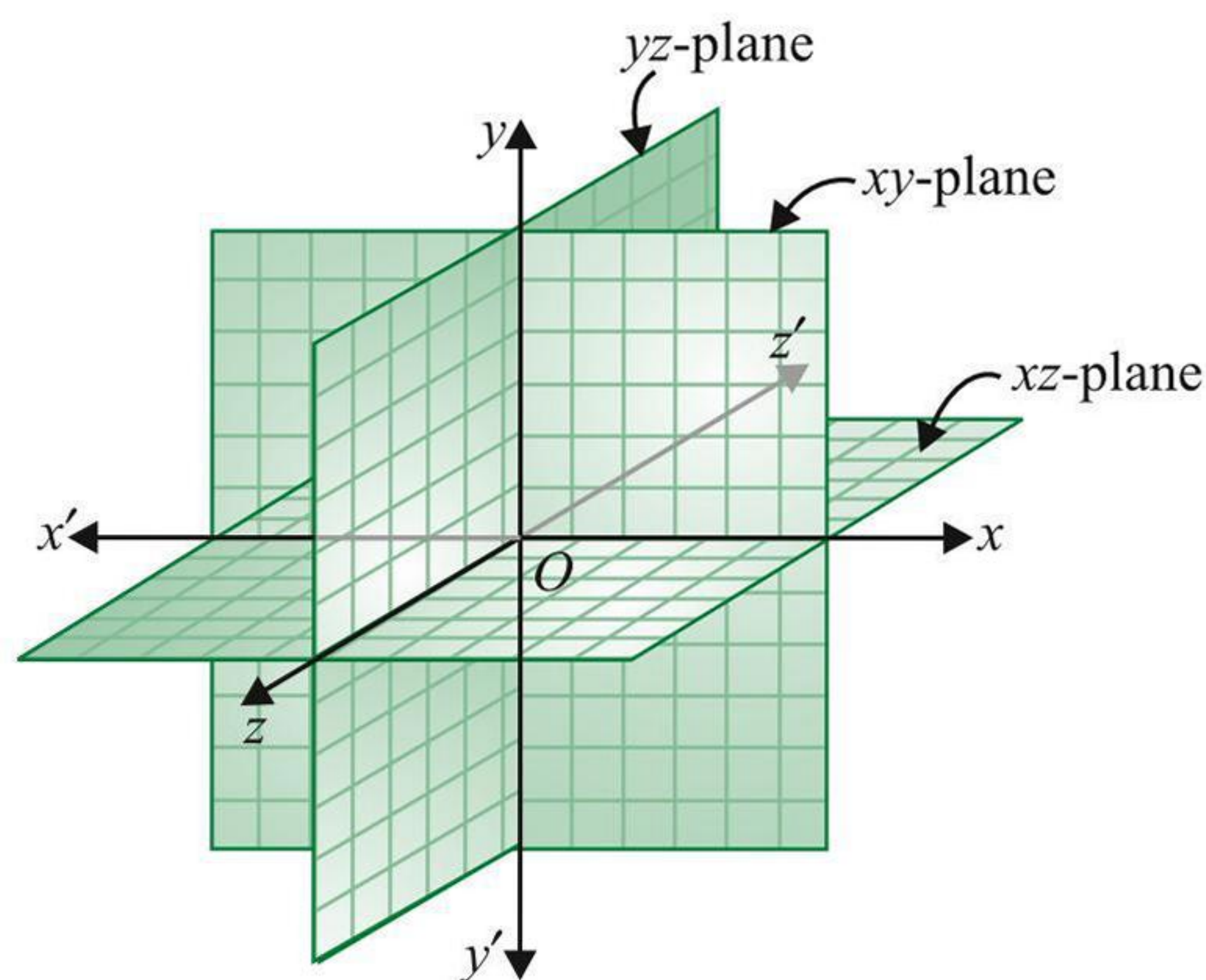
We know that abscissa x of point P is signed distance of P from y -axis i.e., signed distance of P measured parallel to x -axis. Similarly, ordinate y of point P is the signed distance of P from x -axis i.e., signed distance of P measured parallel to y -axis.

Here, two axes divide the plane into four quadrants.

Points on this plane represent ordered pairs of set of cartesian product $R \times R$, where R is set of real numbers.

In three-dimensional coordinate system, we insert one more axis, called z -axis, which goes through the origin of the plane formed by x -axis and y -axis and is perpendicular to both of these axes. Thus, three axes, x -axis, y -axis and z -axis are mutually perpendicular which defines space or three-dimensional (3D) coordinate system. Any point in this coordinate system has coordinates (x, y, z) . Points in this system represent ordered triplets of set of cartesian product $R \times R \times R$, where R is set of real numbers. Three axes taken two at a time form three mutually perpendicular planes, xy -plane, yz -plane and xz -plane. These planes divide space in eight regions called **octants**.

Here Ox , Oy and Oz are positive axes and Ox' , Oy' and Oz' are corresponding negative axes.



The sign of the coordinates of a point determine the octant in which the point lies. The following table shows the signs of the coordinates in eight octants.

Octant \ Coordinates	I	II	III	IV	V	VI	VII	VIII
x	+	−	−	+	+	−	−	+
y	+	+	−	−	+	+	−	−
z	+	+	+	+	−	−	−	−

ILLUSTRATION 1.1

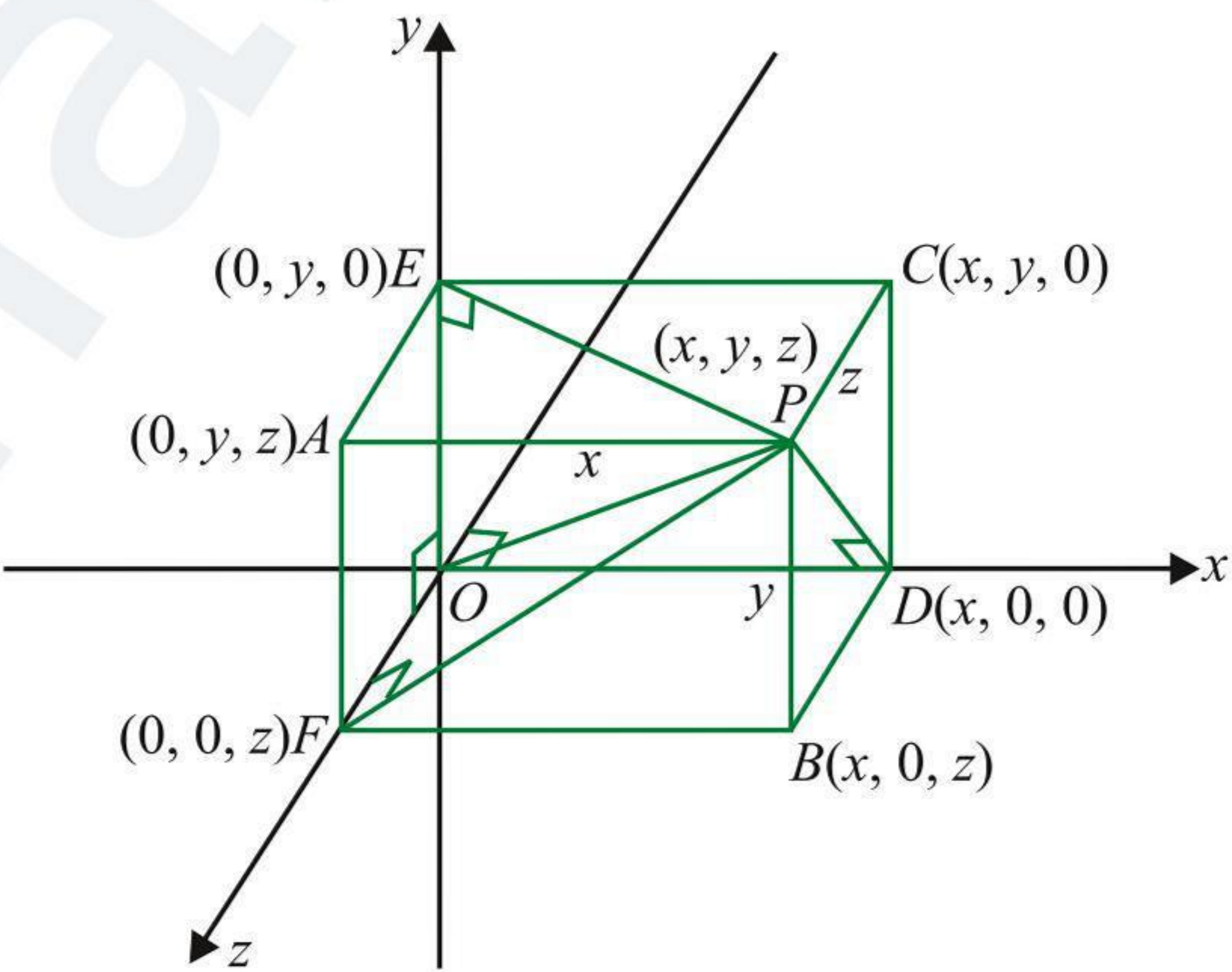
Name the octants in which the following points lie:
(2, 3, 4), (5, −3, 4), (2, −3, −7), (8, 6, −3), (−3, 5, −8), (−9, 3, 4), (−2, −5, 8), (−7, −8, −1)

Sol.

Point	Octant
(+2, +3, +4)	I
(+5, −3, +4)	IV
(+2, −3, −7)	VIII
(+8, +6, −3)	V
(−3, +5, −8)	VI
(−9, +3, +4)	II
(−2, −5, +8)	III
(−7, −8, −1)	VII

UNDERSTANDING COORDINATES OF POINT IN 3D SYSTEM

Consider point $P(x, y, z)$ in 3D coordinate system.



The x -coordinates of point P is its signed distance (PA) from yz -plane, i.e., signed distance of P measured parallel to x -axis.

The y -coordinates of point P is its signed distance (PB) from xz -plane, i.e., signed distance of P measured parallel to y -axis.

The z -coordinates of point P is its signed distance (PC) from xy -plane, i.e., signed distance of P measured parallel to z -axis.

Also, from point P , perpendiculars PD , PE and PF are drawn to x -axis, y -axis and z -axis, respectively.

So, $OD = x$ (with sign). Therefore, coordinates of point D are $(x, 0, 0)$.

Also, OD is projection of OP on x -axis.

Similarly, $OE = y$ (with sign) and $OF = z$ (with sign).

So, points E and F have coordinates $(0, y, 0)$ and $(0, 0, z)$, respectively.

OE and OF is projection of OP on y -axis and z -axis, respectively.

Foot of perpendicular from point P on xz -plane is B .

Clearly, $BD = OF = z$ and $BF = OD = x$.

Also, y -coordinate of any point on xz -plane is zero since point has zero distance measured parallel to y -axis.

So, coordinates of point B are $(x, 0, z)$.

Similarly, coordinates of point A (on yz -plane) are $(0, y, z)$ and those of point C (on xy -plane) are $(x, y, 0)$.

Since x -coordinate of all the points lying on yz -plane is zero, we can say that points on yz -plane satisfy equation $x = 0$, which is equation of yz -plane. In other words, locus of point for which $x = 0$ is yz -plane.

Similarly, equation of xz -plane and xy -plane is $y = 0$ and $z = 0$ respectively.

Equations $y = 0$ and $z = 0$ together is equation of x -axis.

Similarly, equation of y -axis is $x = 0, z = 0$ together and that of z -axis is $x = 0, y = 0$ together.

Distances of Point P from Axes

In right angled triangle PBD , $l(PB) = |y|$ and $l(BD) = l(PC) = |z|$.

$$\begin{aligned} \text{So, } (l(PD))^2 &= (l(PB))^2 + (l(PC))^2 \\ &= (|y|)^2 + (|z|)^2 \\ &= y^2 + z^2 \end{aligned}$$

$$\therefore l(PD) = \sqrt{y^2 + z^2}$$

This is the distance of point P from x -axis.

Similarly, the distance of point P from y -axis is $l(PE) = \sqrt{x^2 + z^2}$ and that of from z -axis is

$$l(PF) = \sqrt{x^2 + y^2}.$$

Distances of Point P from Origin

In right angled triangle ODP ,

$$\begin{aligned} (l(OP))^2 &= (l(OD))^2 + (l(DP))^2 \\ &= (|x|)^2 + (\sqrt{y^2 + z^2})^2 \\ &= x^2 + y^2 + z^2 \end{aligned}$$

$$\therefore l(OP) = \sqrt{x^2 + y^2 + z^2}$$

ILLUSTRATION 1.2

Let A , B and C be the feet of perpendiculars from a point $P(-4, 5, 9)$ on the x , y , z -axes, respectively. Find the coordinates of A , B and C . Also, if the feet of perpendiculars from P on the xy , yz and zx plane are L , M , N , respectively, then find the coordinates of L , M and N .

Sol. We know that on x -axis, $y, z = 0$, on y -axis, $x, z = 0$ and on z -axis, $x, y = 0$.
Thus, the feet of perpendiculars from given point $P(-4, 5, 9)$ on the axis are $A(-4, 0, 0)$, $B(0, 5, 0)$ and $C(0, 0, 9)$.
We know that, on xy -plane $z = 0$, on yz -plane, $x = 0$ and on zx -plane, $y = 0$.
Thus, the coordinates of feet of perpendicular on the xy , yz and zx plane from the given point are $L(-4, 5, 0)$, $M(0, 5, 9)$ and $N(-4, 0, 9)$.

ILLUSTRATION 1.3

Find the distance of point $P(3, -5, 6)$ from x , y and z -axes and from xy , yz and zx -planes.

Sol. Distance of point $P(3, -5, 6)$ from x -axis

$$= \sqrt{y^2 + z^2} = \sqrt{(-5)^2 + 6^2} = \sqrt{61}$$

Distance of point $P(3, -5, 6)$ from y -axis

$$= \sqrt{x^2 + z^2} = \sqrt{3^2 + 6^2} = \sqrt{45}$$

Distance of point $P(3, -5, 6)$ from z -axis

$$= \sqrt{x^2 + y^2} = \sqrt{3^2 + (-5)^2} = \sqrt{34}$$

Distance of point $P(3, -5, 6)$ from xy -plane $= |z| = 6$

Distance of point $P(3, -5, 6)$ from yz -plane $= |x| = 3$

Distance of point $P(3, -5, 6)$ from zx -plane $= |y| = |-5| = 5$

ILLUSTRATION 1.4

Point $P(2, -3, 4)$ is closest to which coordinate plane and which axis?

Sol. Distance of point $P(2, -3, 4)$ from coordinate planes and coordinate axes is as follows.

	Distance
From xy -plane	$ z = 4$
From yz -plane	$ x = 2$
From zx -plane	$ y = 3$
From x -axis	$\sqrt{y^2 + z^2} = \sqrt{(-3)^2 + 4^2} = 5$
From y -axis	$\sqrt{x^2 + z^2} = \sqrt{2^2 + 4^2} = \sqrt{20}$
From z -axis	$\sqrt{x^2 + y^2} = \sqrt{2^2 + (-3)^2} = \sqrt{13}$

Thus, point P is closest to yz -plane and to z -axis.

ILLUSTRATION 1.5

Find the reflection point $P(2, 3, -4)$ across xy , yz and zx -plane.

Sol. Given point is $P(2, 3, -4)$.

If image of point P in xy -plane is Q then PQ will be parallel to z -axis.

So, x and y -coordinate of point Q will be same and z -coordinate will change its sign.

Hence $Q \equiv (2, 3, 4)$

With similar reasoning, image of P in yz -plane is $R(-2, 3, -4)$ and in zx -plane is $S(2, -3, -4)$.

ILLUSTRATION 1.6

If distance of point P from x , y and z -axis is 2, 3, 6 then find its distance from origin.

Sol. Let $P \equiv (x, y, z)$

Its distance from x , y and z axis is

$$\sqrt{y^2 + z^2} = 2,$$

$$\sqrt{x^2 + z^2} = 3,$$

and $\sqrt{x^2 + y^2} = 6$

Squaring and adding, we get $2(x^2 + y^2 + z^2) = (2^2 + 3^2 + 6^2) = 49$

$$\therefore x^2 + y^2 + z^2 = 49/2$$

Hence, $\sqrt{x^2 + y^2 + z^2} = \frac{7}{\sqrt{2}}$, which is distance of point P from origin.

DIRECTION ANGLES AND DIRECTION COSINES

In the figure, let OP make positive angles (anticlockwise direction) α , β and γ with the positive x , y and z -axes, respectively.

These angles are known as direction angles.

Taking cosine of these angles we get direction cosines, viz., $\cos \alpha$, $\cos \beta$, $\cos \gamma$.

Direction cosines makes it easy to represent the direction of a line and vector in space.

In the figure, in triangle ODP ,

$$\cos \alpha = \frac{OD}{OP} = \frac{x}{\sqrt{x^2 + y^2 + z^2}} = l$$

Similarly, in triangle OEP ,

$$\cos \beta = \frac{OE}{OP} = \frac{y}{\sqrt{x^2 + y^2 + z^2}} = m$$

and in triangle OFP ,

$$\cos \gamma = \frac{OF}{OP} = \frac{z}{\sqrt{x^2 + y^2 + z^2}} = n$$

Thus, $l^2 + m^2 + n^2 = 1$.

This is the relation between the direction cosines.

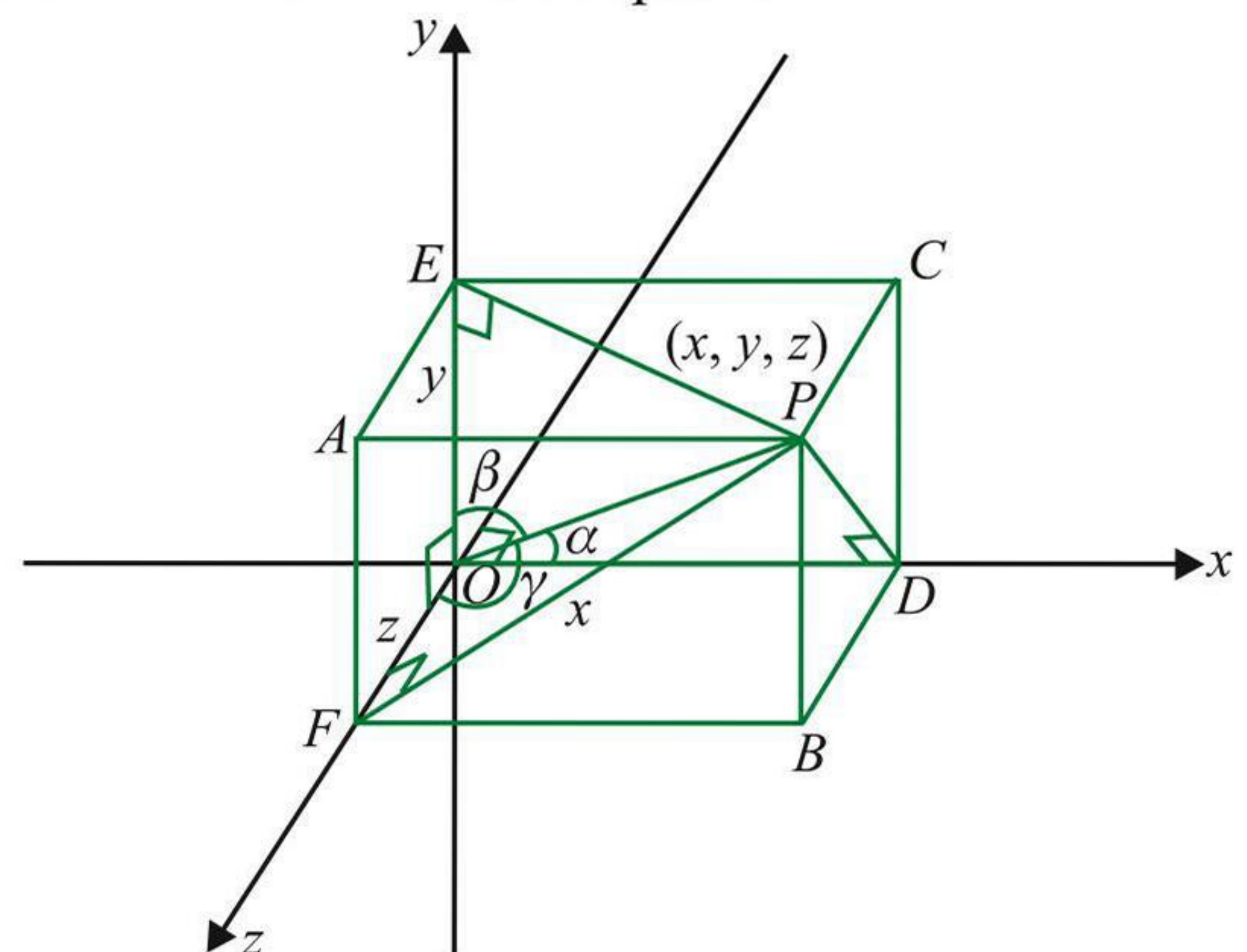


ILLUSTRATION 1.7

If α , β and γ are the angle which a directed line makes with the positive directions of the coordinate axes, then find the value of $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma$.

Sol. The DCs of the line are $l = \cos \alpha$, $m = \cos \beta$, $n = \cos \gamma$

We know that

$$l^2 + m^2 + n^2 = 1$$

$$\therefore \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\Rightarrow 1 - \sin^2 \alpha + 1 - \sin^2 \beta + 1 - \sin^2 \gamma = 1$$

$$\Rightarrow \sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$$

ILLUSTRATION 1.8

A line OP through origin O is inclined at 30° and 45° to OX and OY , respectively. Then find the angle at which it is inclined to OZ .

Sol. Let l , m , n be the direction cosines of OP , then $l^2 + m^2 + n^2 = 1$.

$$\text{If } l = \cos 30^\circ = \frac{\sqrt{3}}{2}, m = \cos 45^\circ = \frac{1}{\sqrt{2}}, \text{ then } \frac{3}{4} + \frac{1}{2} + n^2 = 1$$

$$\Rightarrow n^2 = -\frac{1}{4}, \text{ which is not possible.}$$

So, such a line cannot exist.

ILLUSTRATION 1.9

A line OP makes equal angle with coordinate axes then find its direction cosines.

Sol. Let l , m , n be the direction cosines of OP , which makes equal angle α with coordinate axes.

$$\therefore l = m = n = \cos \alpha$$

$$\text{Now } l^2 + m^2 + n^2 = 1$$

$$\therefore 3l^2 = 1$$

$$\Rightarrow l = \pm \frac{1}{\sqrt{3}} = m = n$$

$$\text{Therefore, } l = m = n = \frac{1}{\sqrt{3}} \text{ or } l = m = n = -\frac{1}{\sqrt{3}}$$

For $l = m = n = \frac{1}{\sqrt{3}}$, point P lies in first octant.

For $l = m = n = -\frac{1}{\sqrt{3}}$, point P lies in seventh octant, where OP makes equal angle $(180^\circ - \alpha)$ with coordinate axes.

ILLUSTRATION 1.10

OP has direction cosines, $\cos \alpha$, $\cos \beta$ and $\cos \gamma$. Assuming yz -plane as a mirror then what are the direction cosines of the reflected image of OP in the yz -plane.

Sol. Let point P lies in first octant.

$\therefore$ Directional angles of OP are α , β and γ .

If OP is reflected in yz -plane and OP' is reflected line segment then OP' will make same angle β and γ with y -axis and z -axis, respectively, but its inclination with x -axis will be $(180^\circ - \alpha)$

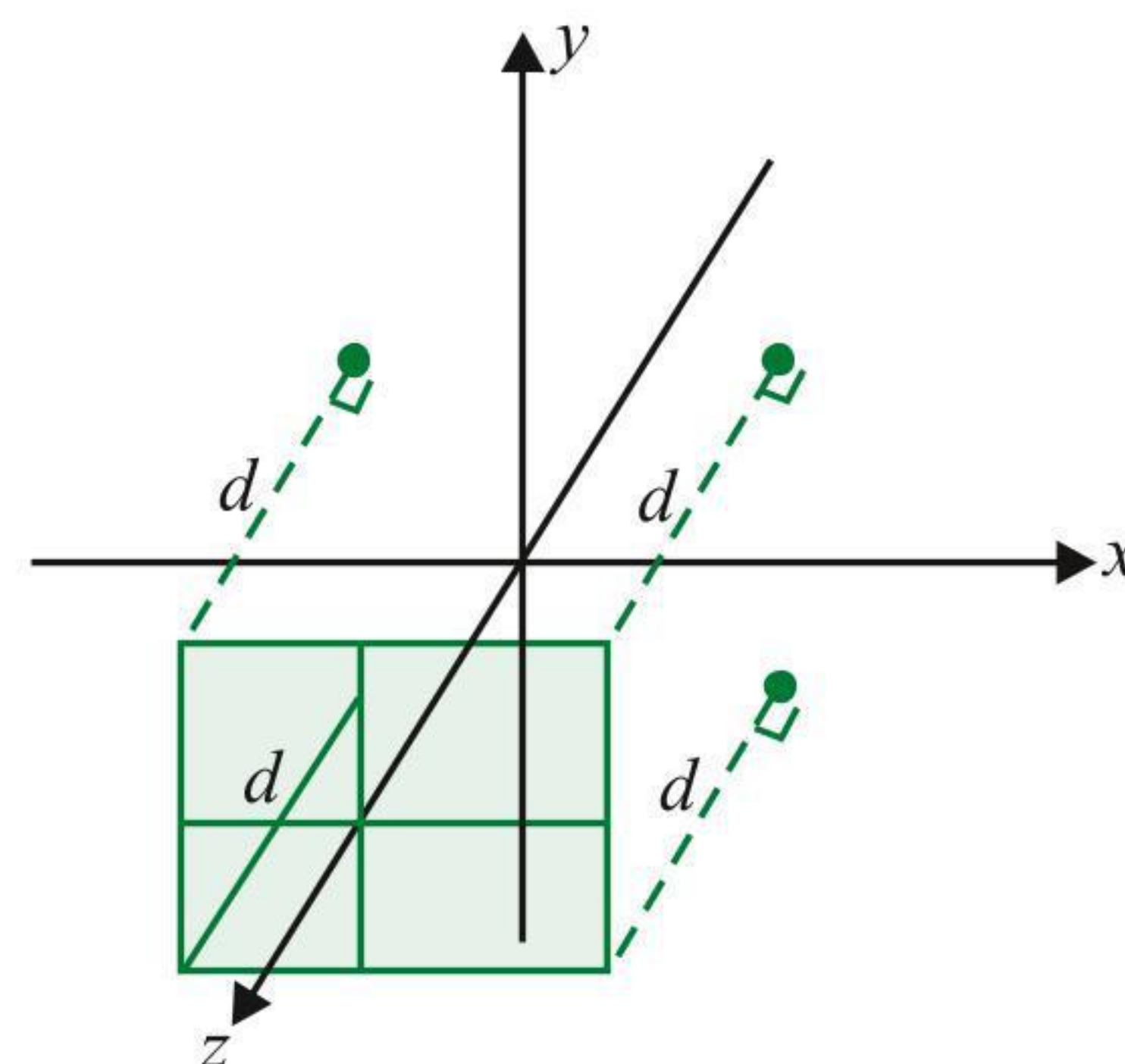
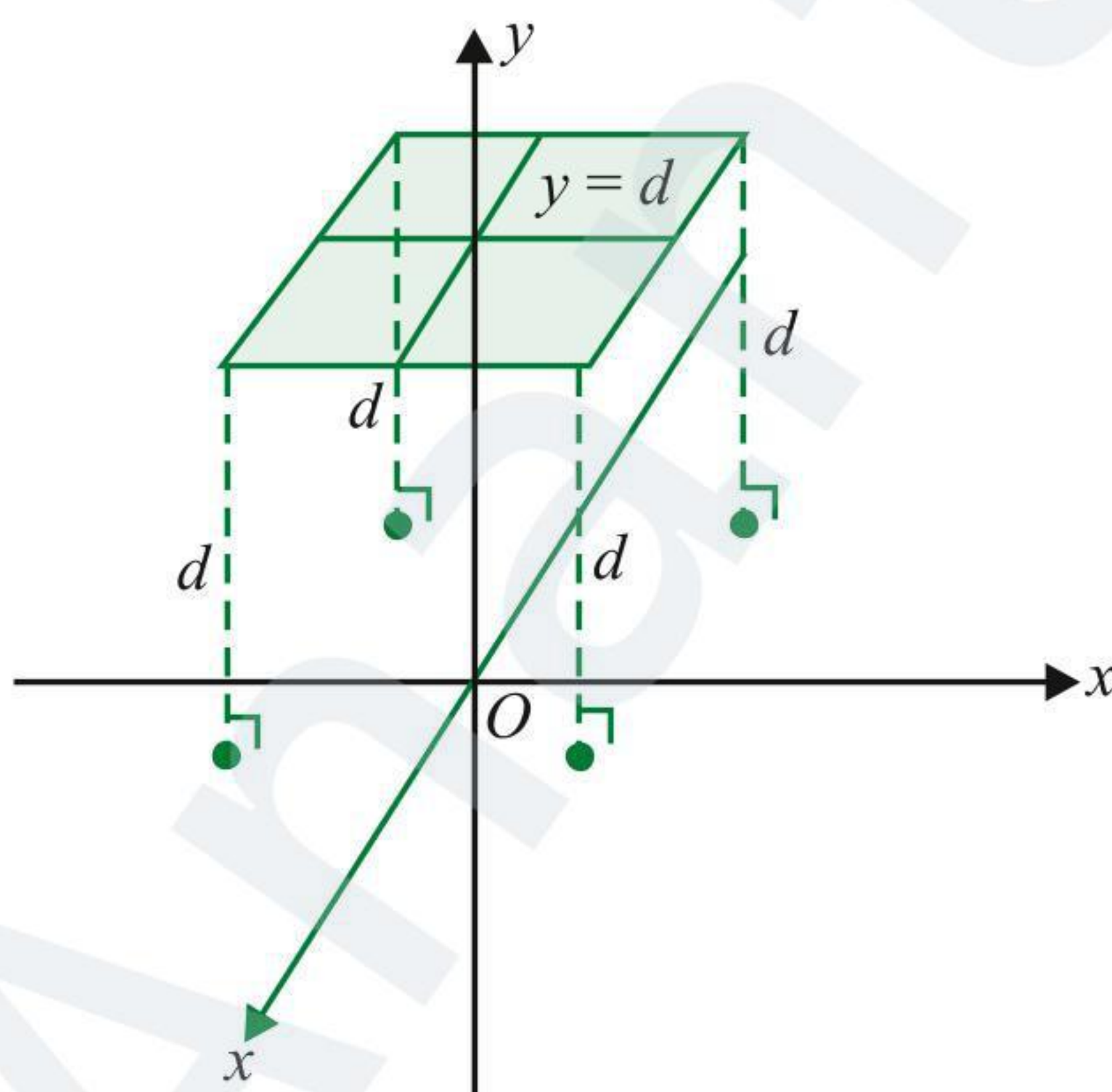
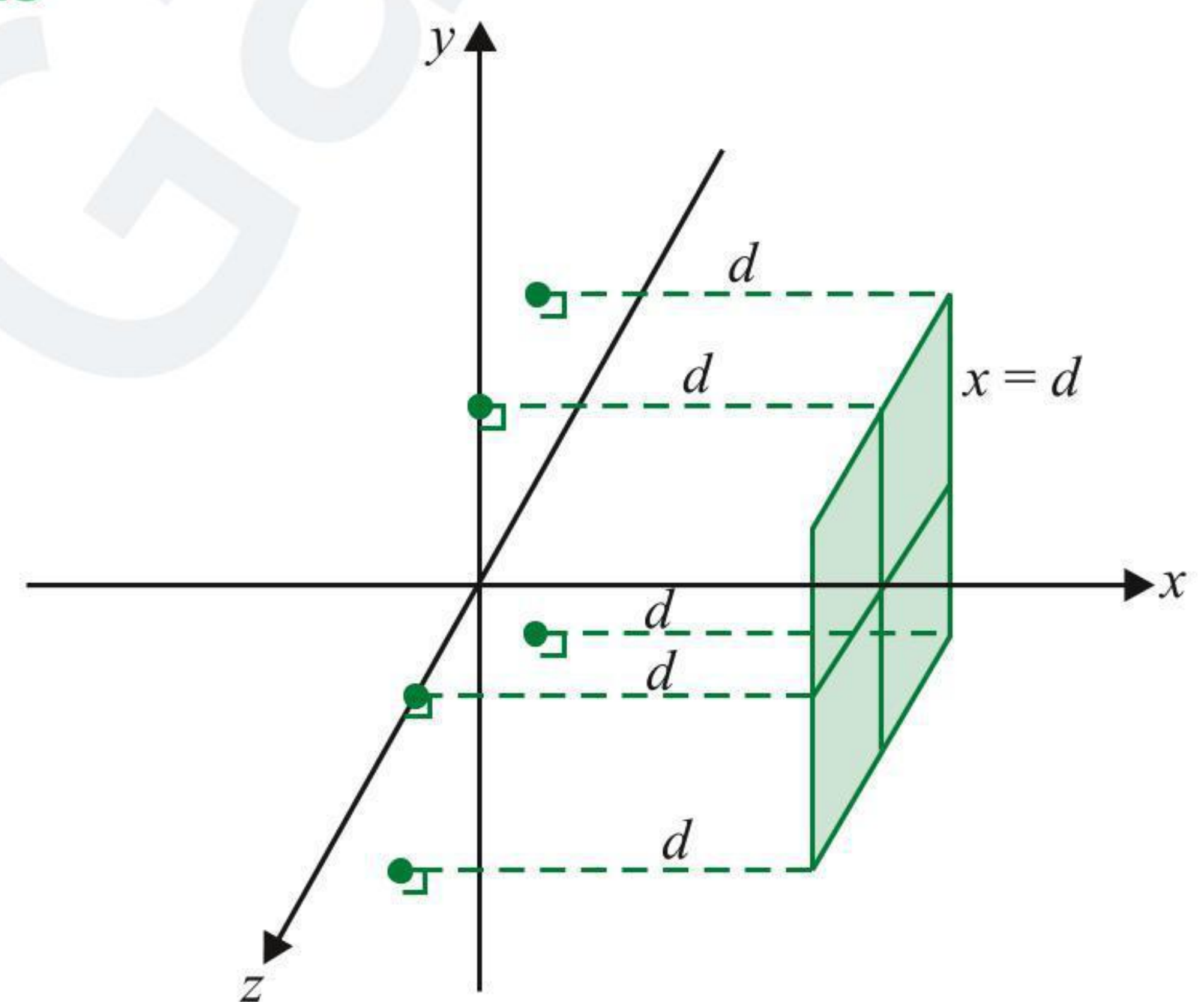
So, direction cosines of OP' are $\cos(180^\circ - \alpha)$, $\cos \beta$, $\cos \gamma$ or $-\cos \alpha$, $\cos \beta$ and $\cos \gamma$.

PLANE PERPENDICULAR TO ONE OF THE THREE AXES

We know that the signed distance of point $P(x_1, y_1, z_1)$ from yz -plane is x_1 . If we keep x_1 constant and change the values of y_1 and z_1 arbitrarily, the signed distance of different points thus obtained from yz -plane will remain constant x_1 . All these points form a plane parallel to yz -plane.

Thus, locus of the point whose signed distance from yz -plane is constant d is $x = d$. This is equation of plane parallel to yz -plane whose signed distance from yz -plane is d .

Similarly, equation of plane whose signed distance from xz -plane is d is $y = d$ and that of plane whose signed distance from xy -plane d is $z = d$.



Equations of planes at distance d from the yz -plane are $x = \pm d$. Similarly, equations of planes at distance d from xz -plane are $y = \pm d$ and those at distance d from xy -plane are $z = \pm d$.

ILLUSTRATION 1.11

Find the equations of planes passing through point $(-2, 3, -5)$ and which are parallel to

- (i) xy -plane (ii) yz -plane (iii) zx -plane

Sol.

- (i) Signed distance of point $P(-2, 3, -5)$ from xy -plane is -5 .
So, equation of plane through point P and parallel to xy -plane is $z = -5$.
- (ii) Signed distance of point $P(-2, 3, -5)$ from xz -plane is 3 .
So, equation of plane through point P and parallel to xz -plane is $y = 3$.
- (iii) Signed distance of point $P(-2, 3, -5)$ from yz -plane is -2 .
So, equation of plane through point P and parallel to yz -plane is $x = -2$.

LINE PARALLEL TO ONE OF THE THREE AXES

Two planes in a space intersect on a line.

For example, xy -plane and yz -plane intersect on y -axis whose equation is $x = 0, z = 0$ together.

Now, consider planes $x = d_1$ which is parallel to yz -plane and $z = d_2$, which is parallel to xy -plane.

These two planes intersect on a line L which is parallel to y -axis.

Equation of this line is $x = d_1, z = d_2$ together and lies at distance $\sqrt{d_1^2 + d_2^2}$ from y -axis.

Any point on this line has x -coordinate d_1 and z -coordinate d_2 .
For different points on the line y -coordinate will be different.

In the figure, $OA = d_1$ and $AC = OB = d_2$.

In triangle OAC , $OC^2 = OA^2 + AC^2 = d_1^2 + d_2^2$

$$\therefore OC = \sqrt{d_1^2 + d_2^2}.$$

Similarly, $x = d_1, y = d_3$ together is equation of line parallel to z -axis at distance $\sqrt{d_1^2 + d_3^2}$ from it and $y = d_2, z = d_3$ together is equation of line parallel to x -axis at distance $\sqrt{d_2^2 + d_3^2}$ from it.

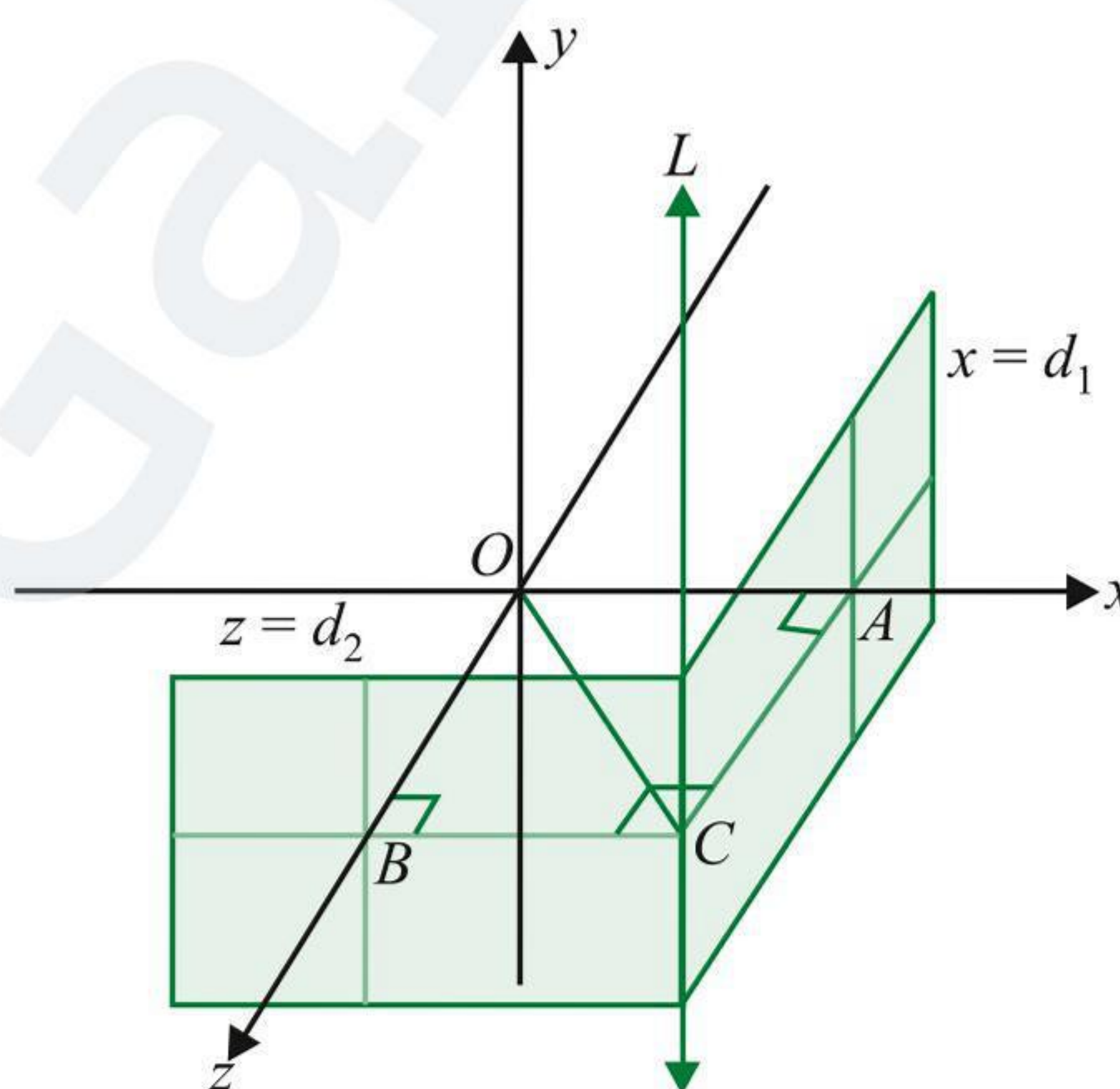


ILLUSTRATION 1.12

What is the distance of line $x = 3, y = 4$ from axis to which it is parallel?

Sol. Clearly, line $x = 3, y = 4$ is parallel to z -axis.

Its distance from z -axis is $\sqrt{3^2 + 4^2} = 5$

ILLUSTRATION 1.13

Find the equation of line through point $P(2, 3, -4)$, which is

- (i) parallel to x -axis (ii) parallel to y -axis (iii) parallel to z -axis.

Sol. Given point is $P(2, 3, -4)$

Now equation of planes through point P which are parallel to yz -plane, xz -plane and xy -plane are $x = 2$, $y = 3$ and $z = -4$, respectively.

- (i) equation of line through point P which is parallel to x -axis is $y = 3, z = -4$
- (ii) equation of line through point P which is parallel to y -axis is $x = 2, z = -4$
- (iii) equation of line through point P which is parallel to z -axis is $x = 2, y = 3$

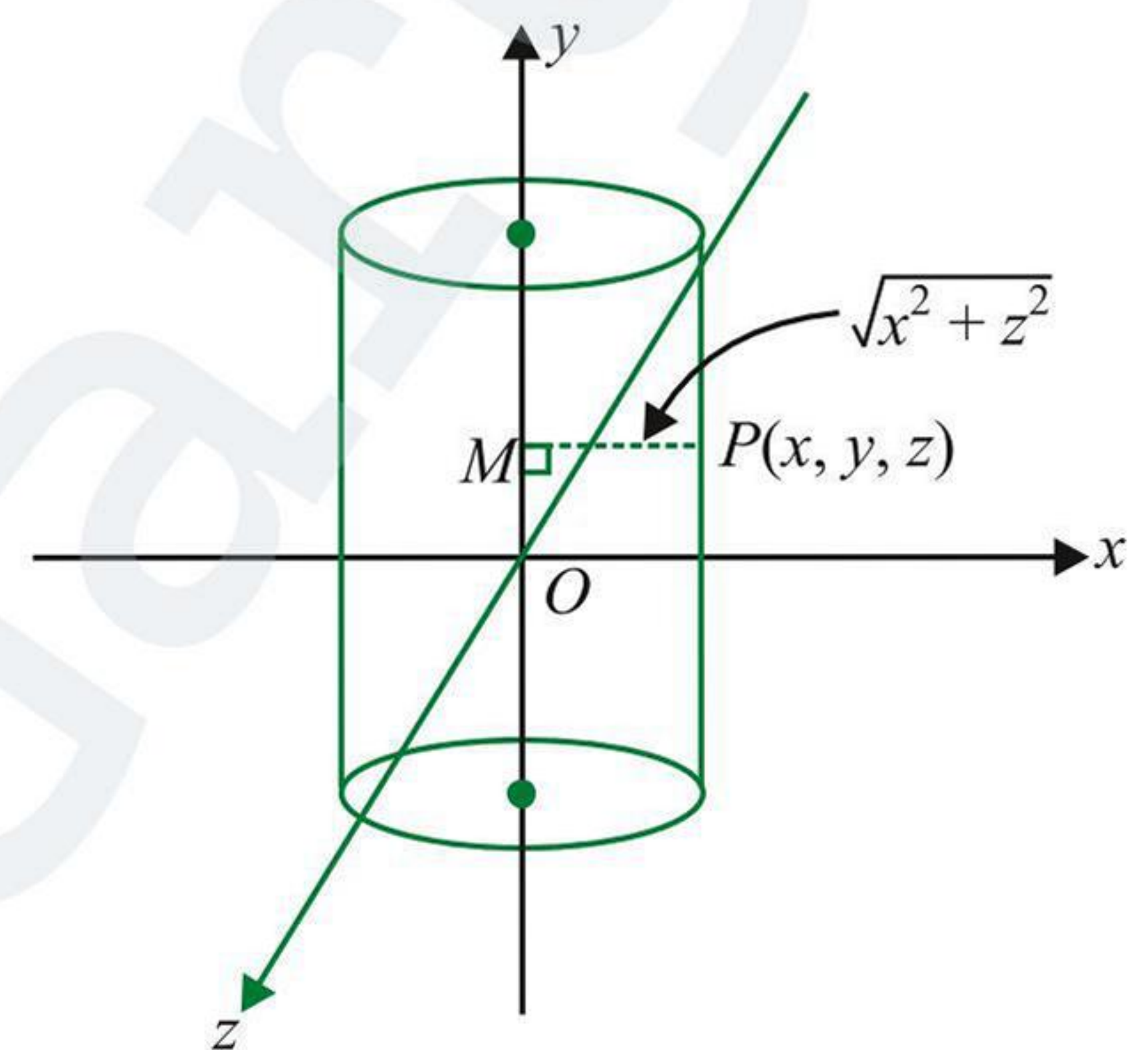
ILLUSTRATION 1.14

Explain the locus whose equation is $x^2 + z^2 = 25$.

Sol. Let variable point $P(x, y, z)$ satisfy the equation $x^2 + z^2 = 25$.

$\therefore \sqrt{x^2 + z^2} = 5$, i.e., distance of point P from y -axis is constant '5'.

Clearly, all such points lie on the curved surface of cylinder whose axis is y -axis and radius is 5.

**ILLUSTRATION 1.15**

Explain equation $y = 3, z = 4$

Sol. Equation $y = 3$ represents plane at distance 3 units on cutting positive side of y -axis.

Equation $z = 4$ represents plane at distance 4 units on cutting positive side of z -axis.

Equations $y = 3, z = 4$ together represents points on the line of intersection of planes $y = 3$ and $z = 4$ which is parallel to x -axis and at distance $\sqrt{3^2 + 4^2} = 5$ from x -axis.

CONCEPT APPLICATION EXERCISE 1.1

- A parallelepiped is formed by planes drawn through the points $P(6, 8, 10)$ and $Q(3, 4, 8)$ parallel to the coordinate planes. Find the length of an edge of the parallelepiped.
- What does the equation $x^2 - 2x - 3 = 0$ represent in 3D coordinate system?
- What is distance of point $P(3, 4, -5)$ from
 - (i) plane $z = 6$
 - (ii) plane $x = 7$
- Find the image of point $P(-3, 2, 4)$ in xy , yz and zx -plane and also find the octant in which they lie.
- A line makes angles α, β and γ with the coordinate axes. If $\alpha + \beta = 90^\circ$, then find γ .
- If a line makes the angles α, β and γ with 3D coordinate axes, then find the value of $\cos 2\alpha + \cos 2\beta + \cos 2\gamma$.
- Let point P be $P(2, 3, 6)$ and O be the origin. Then with which axis the angle of OP is the greatest?

ANSWERS

- | | | |
|------------------------|------------------------------------|------------------------------|
| 1. $\sqrt{39}$ | 2. two planes $x = -1$ and $x = 3$ | 3. (i) 11 units (ii) 4 units |
| 5. $\gamma = 90^\circ$ | 6. -1 | 7. x -axis |

DISTANCE FORMULA

DISTANCE BETWEEN TWO POINTS LYING ON THE SAME AXIS

Consider two points $P_1(x_1, 0, 0)$ and $P_2(x_2, 0, 0)$ on x -axis.

So, distance between these two points is $P_1P_2 = |x_1 - x_2|$.

Similarly, distance between two points $Q_1(0, y_1, 0)$ and $Q_2(0, y_2, 0)$ on y -axis is $|y_1 - y_2|$ and distance between two points $R_1(0, 0, z_1)$ and $R_2(0, 0, z_2)$ on z -axis is $|z_1 - z_2|$.

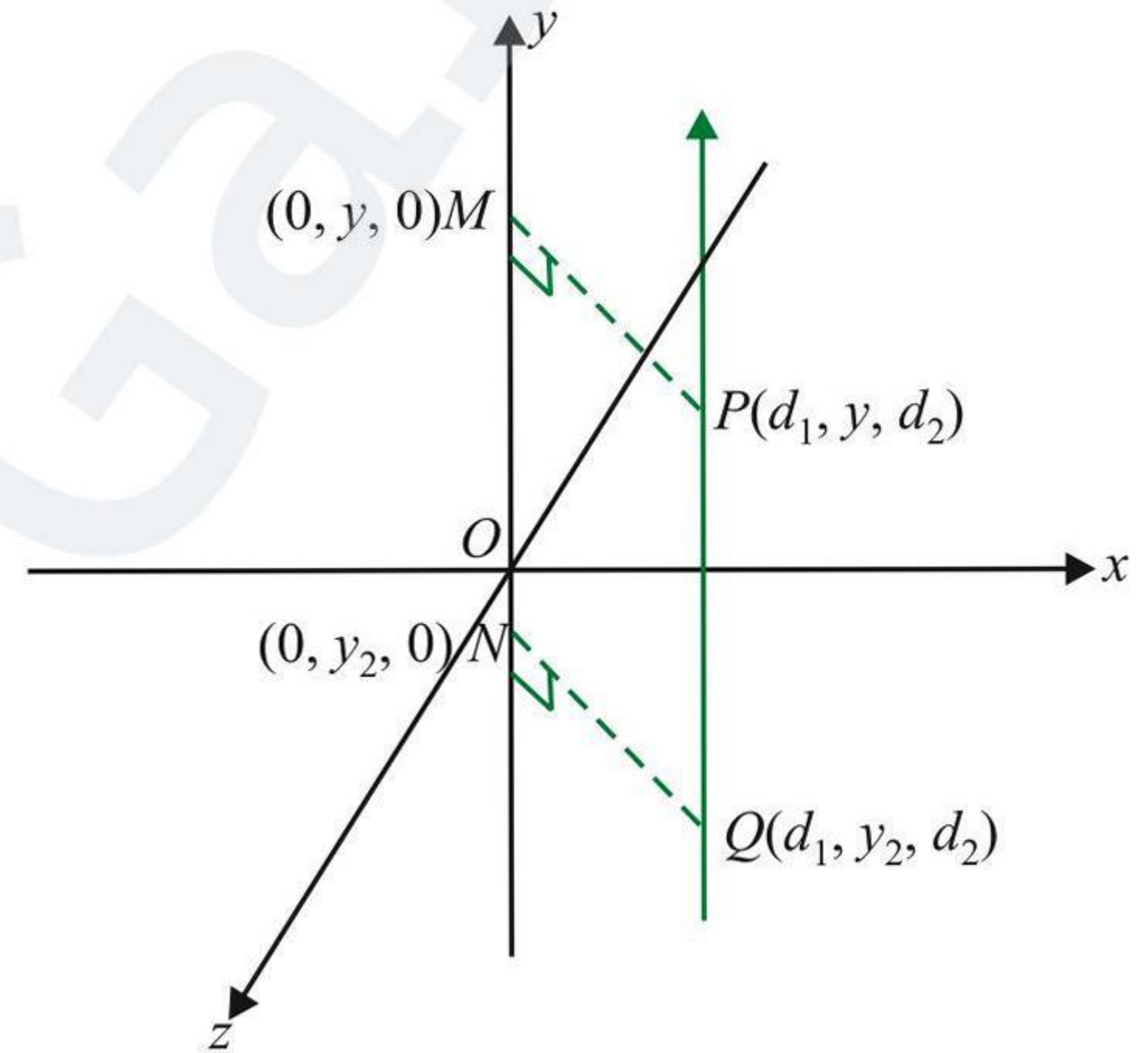
DISTANCE BETWEEN TWO POINTS LYING ON THE LINE PARALLEL TO ONE OF THE AXES

Consider line having equation $x = d_1, z = d_2$. Clearly, this line is parallel to y -axis. Any point on this line has x -coordinate d_1 and z -coordinate d_2 . Consider two points on the line $P(d_1, y_1, d_2)$ and $Q(d_1, y_2, d_2)$.

Projection of points P and Q on y -axis are $M(0, y_1, 0)$ and $N(0, y_2, 0)$, respectively.

So, $PQ = MN = |y_1 - y_2|$

Similarly, distance between two points on the line parallel to x -axis is difference between their x -coordinates and distance between two points on the line parallel to z -axis is difference between their z -coordinates.



DISTANCE BETWEEN TWO POINTS

Consider two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$. Cuboid is constructed with PQ as longest diagonal. PA is parallel to x -axis.

Vertex A is foot of perpendicular from vertex Q on side PA . So, coordinates of point A are (x_2, y_1, z_1) . Therefore, $PA = |x_1 - x_2|$.

Similarly, $PB = |y_1 - y_2|$ and $PC = |z_1 - z_2|$.

In right angled triangle PAB' ,

$$\begin{aligned} PB'^2 &= PA^2 + AB'^2 \\ &= PA^2 + PC^2 \\ &= (x_1 - x_2)^2 + (z_1 - z_2)^2 \end{aligned}$$

Now, in right angled triangle $PB'Q$,

$$\begin{aligned} PQ^2 &= PB'^2 + B'Q^2 \\ &= PB'^2 + PB^2 \\ &= (x_1 - x_2)^2 + (z_1 - z_2)^2 + (y_1 - y_2)^2 \end{aligned}$$

$$\therefore PQ = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

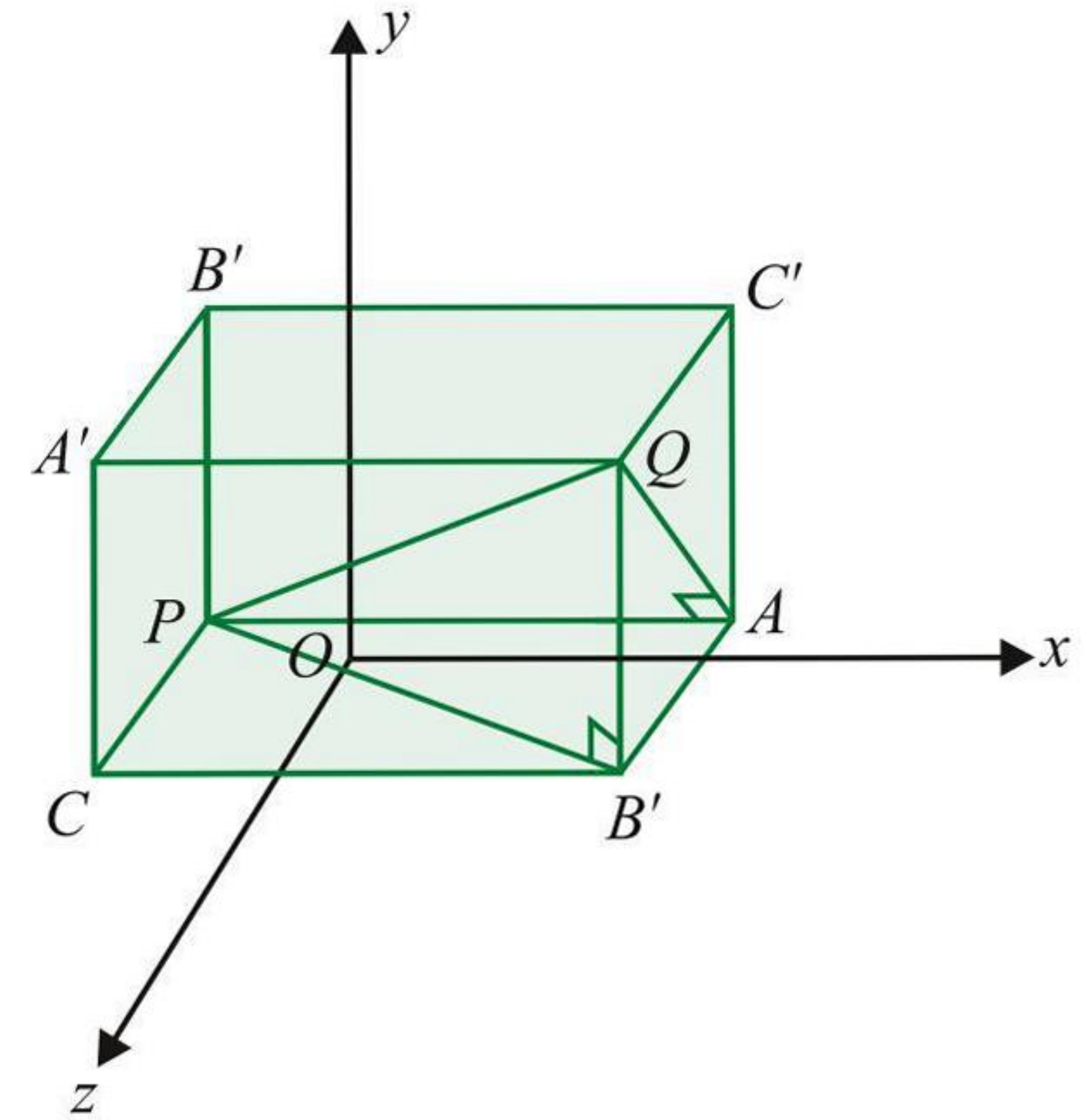


ILLUSTRATION 1.16

Given points $A(-2, 1, 4)$ and $B(5, 0, -7)$. Find the distance between A and B .

Sol. Distance between A and B

$$\begin{aligned} &= \sqrt{(5 - (-2))^2 + (0 - 1)^2 + (-7 - 4)^2} \\ &= \sqrt{49 + 1 + 121} = \sqrt{171} \end{aligned}$$

ILLUSTRATION 1.17

Show that the points $A(1, 0, 2)$, $B(-2, 3, 1)$ and $C(0, 4, -2)$ are the vertices of a right angled triangle.

Sol.

$$\begin{aligned} AB &= \sqrt{(1 - (-2))^2 + (0 - 3)^2 + (2 - 1)^2} = \sqrt{9 + 9 + 1} = \sqrt{19} \\ AC &= \sqrt{(1 - 0)^2 + (0 - 4)^2 + (2 - (-2))^2} = \sqrt{1 + 16 + 16} = \sqrt{33} \\ BC &= \sqrt{(-2 - 0)^2 + (3 - 4)^2 + (1 - (-2))^2} = \sqrt{4 + 1 + 9} = \sqrt{14} \end{aligned}$$

Clearly, $AC^2 = AB^2 + BC^2$

So, triangle ABC is right angled triangle with right angle at vertex B .

ILLUSTRATION 1.18

Show that points $A(2, 3, 5)$, $B(6, 1, 6)$ and $C(14, -3, 8)$ are collinear.

Sol. If A , B and C lie on the same line, then the largest distance between any two of the three points will be equal to the sum of the two smaller distances.

$$\begin{aligned} \text{Now, } AB &= \sqrt{(6 - 2)^2 + (1 - 3)^2 + (6 - 5)^2} \\ &= \sqrt{4^2 + 2^2 + 1^2} = \sqrt{21} \\ AC &= \sqrt{(14 - 2)^2 + (-3 - 3)^2 + (8 - 5)^2} \\ &= \sqrt{12^2 + (-6)^2 + 3^2} = \sqrt{189} = 3\sqrt{21} \\ BC &= \sqrt{(14 - 6)^2 + (-3 - 1)^2 + (8 - 6)^2} \\ &= \sqrt{8^2 + (-4)^2 + 2^2} = \sqrt{84} = 2\sqrt{21} \end{aligned}$$

Hence, $AB + BC = AC$.

Therefore, point A , B and C are collinear with point B dividing AC in the ratio 1 : 2.

ILLUSTRATION 1.19

Find the coordinates of a point on z -axis which are at a distance of 7 units from the point $P(3, -2, 5)$.

Sol. Point on the z -axis is $A(0, 0, z)$.

Given point is $P(3, -2, 5)$.

Also, it is given that $AP = 7$

$$\begin{aligned} \therefore AP^2 &= 50 \\ \Rightarrow (3-0)^2 + (-2-0)^2 + (5-z)^2 &= 49 \\ \Rightarrow 9 + 4 + (5-z)^2 &= 49 \\ \Rightarrow (5-z)^2 &= 36 \\ \Rightarrow 5-z &= \pm 6 \\ \Rightarrow z &= -1 \text{ or } 11 \end{aligned}$$

Thus, the coordinates of the required points are $(0, 0, -1)$ and $(0, 0, 11)$.

ILLUSTRATION 1.20

Given points $A(-2, 1, 4)$ and $B(5, 0, -7)$. Find all the points which are equidistant from A and B . Also, find the locus of these points.

Sol. Let point $P(x, y, z)$ be equidistant from points A and B .

So, $PA = PB$

$$\begin{aligned} \therefore \sqrt{(x-(-2))^2 + (y-1)^2 + (z-4)^2} &= \sqrt{(x-5)^2 + (y-0)^2 + (z-(-7))^2} \\ \Rightarrow (x+2)^2 + (y-1)^2 + (z-4)^2 &= (x-5)^2 + y^2 + (z+7)^2 \\ \Rightarrow x^2 + 4x + 4 + y^2 - 2y + 1 + z^2 - 8z + 16 &= x^2 - 10x + 25 + y^2 + z^2 + 14z + 49 \\ \Rightarrow 14x - 2y - 22z &= 53. \end{aligned}$$

Thus, all points which are equidistant from A and B satisfy the equation $14x - 2y - 22z = 53$.

CONCEPT APPLICATION EXERCISE 1.2

- Verify the following:
 - $(0, 7, -10)$, $(1, 6, -6)$ and $(4, 9, -6)$ are the vertices of an isosceles triangle.
 - $(-1, 2, 1)$, $(1, -2, 5)$, $(4, -7, 8)$ and $(2, -3, 4)$ are the vertices of a parallelogram.
- Distance of point P from origin is 7 units, from x -axis is $\sqrt{13}$ units and from y -axis is $\sqrt{40}$ units. Then find coordinates of point P .
- Let $x^2 + 3y^2 = 3$ be the equation of an ellipse in the xy -plane. If $A(-\sqrt{3}, 0, 0)$ and $B(-\sqrt{3}, 0, 2)$ are two points, then find point P on the ellipse such that $\angle APB = 45^\circ$.
- Explain the locus of point $P(3, y, z)$ whose distance of origin is 5.

ANSWERS

2. $(\pm 6, \pm 3, \pm 2)$ 3. $(0, \pm 1, 0)$

SECTION FORMULA

Consider two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$. Let us find the coordinates of point R which divides PQ in the ratio $m : n$ internally.

Draw perpendiculars PP' , QQ' and RR' to the xz -plane. P' , Q' and R' are feet of these perpendiculars on xz -plane.

Clearly $PP' = y_1$, $QQ' = y_2$.

Let $RR' = y$.

Now through point R draw a line ST parallel to the line $P'Q'$. Line ST will intersect the line PP' externally at the point S and the line QQ' at T .

Clearly, triangles PSR and QTR are similar.

$$\therefore \frac{m\lambda}{n\lambda} = \frac{PR}{QR} = \frac{SP}{QT} = \frac{SP' - PP'}{QQ' - TQ'} = \frac{RR' - PP'}{QQ' - RR'} = \frac{y - y_1}{y_2 - y}$$

Simplifying, we get,

$$y = \frac{my_2 + ny_1}{m + n}$$

Similarly, by drawing perpendiculars to the yz -plane and xy plane, we get

$$x = \frac{mx_2 + nx_1}{m + n} \text{ and } z = \frac{mz_2 + nz_1}{m + n}$$

Hence, the coordinates of the point R which divides the line segment joining two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ internally in the ratio $m : n$ are

$$\left(\frac{mx_2 + nx_1}{m + n}, \frac{my_2 + ny_1}{m + n}, \frac{mz_2 + nz_1}{m + n} \right)$$

If the point R divides PQ externally in the ratio $m : n$, then its coordinates are obtained by replacing n by $-n$ so that coordinates of point R will be

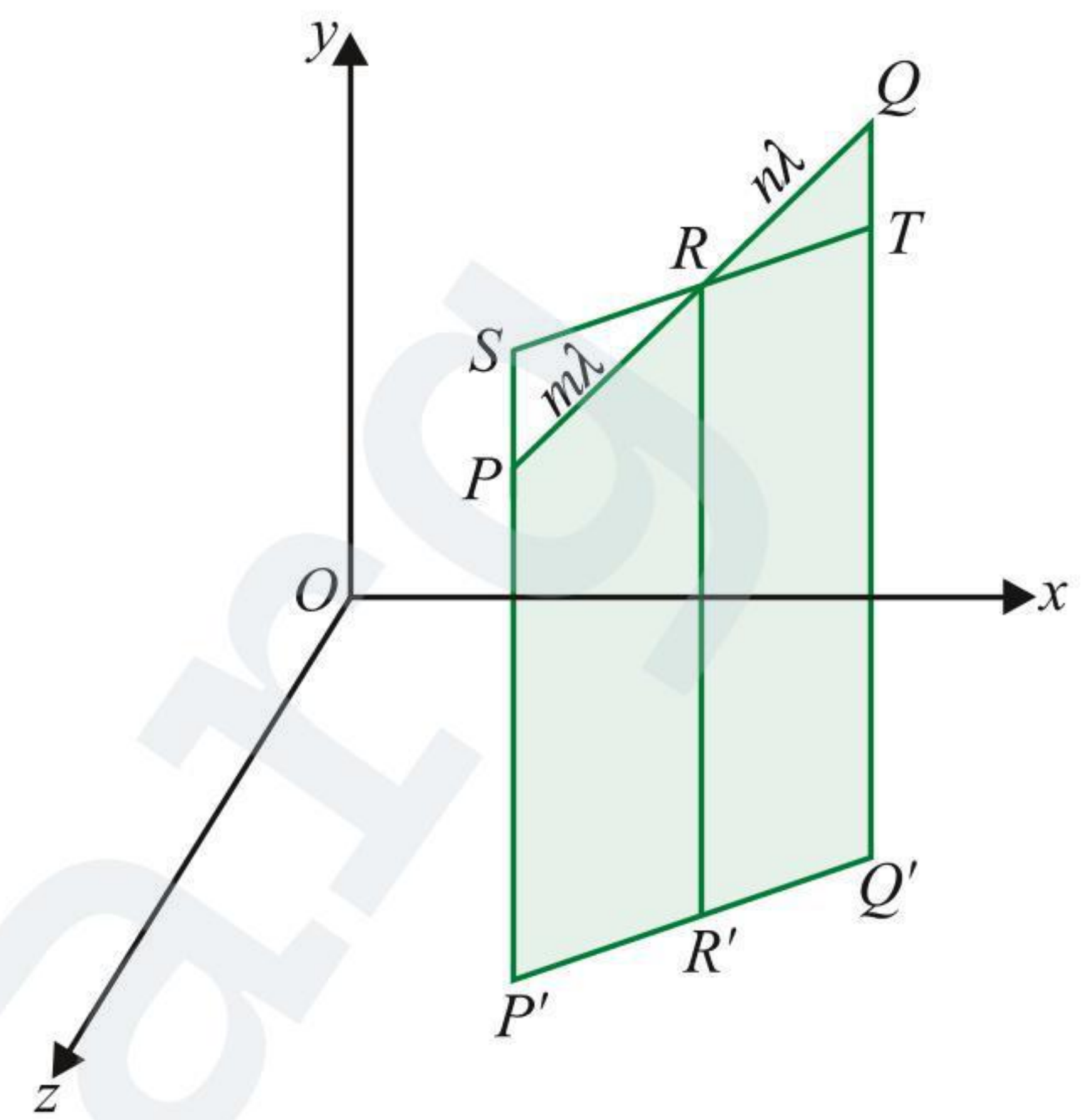
$$\left(\frac{mx_2 - nx_1}{m - n}, \frac{my_2 - ny_1}{m - n}, \frac{mz_2 - nz_1}{m - n} \right)$$

- The coordinates of the mid-point of the line segment joining two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ are $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$.
- The coordinates of the centroid of the triangle, whose vertices are (x_1, y_1, z_1) , (x_2, y_2, z_2) and (x_3, y_3, z_3) are $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3} \right)$.

ILLUSTRATION 1.21

Find the coordinates of the point which divides the line segment joining the points $(-4, 2, 3)$ and $(3, -5, 7)$ in the ratio

- (i) $3 : 4$ internally, and (ii) $3 : 4$ externally.



Sol.

- (i) Let $R(x, y, z)$ be the point that divides the line segment joining points $P(-4, 2, 3)$ and $Q(3, -5, 7)$ internally in the ratio $3 : 4$. Then

$$x = \frac{3(3) + 3(-4)}{3+4}, y = \frac{3(-5) + 4(2)}{3+4} \text{ and } z = \frac{3(7) + 4(3)}{3+4}$$

i.e., $x = -\frac{3}{7}, y = -1, z = \frac{33}{7}$

Thus, $R \equiv \left(-\frac{3}{7}, -1, \frac{33}{7}\right)$.

- (ii) Let $S(x, y, z)$ divides PQ externally in the ratio $3 : 4$ then

$$x = \frac{3(3) - 3(-4)}{3-4}, y = \frac{3(-5) - 4(2)}{3-4} \text{ and } z = \frac{3(7) - 4(3)}{3-4}$$

i.e., $x = -21, y = 23, z = -9$

Thus, $S \equiv (-21, 23, -9)$.

ILLUSTRATION 1.22

Prove that points $P(3, 2, -4)$, $Q(5, 4, -6)$ and $R(9, 8, -10)$ are collinear. Find the ratio in which Q divides PR .

Sol. Let point $Q(5, 4, -6)$ divide the line segment joining points $P(3, 2, -4)$ and $R(9, 8, -10)$ in the ratio $k : 1$. Therefore, by section formula,

$$(5, 4, -6) \equiv \left(\frac{k(9) + 3}{k+1}, \frac{k(8) + 2}{k+1}, \frac{k(-10) - 4}{k+1} \right)$$

If points P , Q and R are collinear, we must find same value of k from each comparison.

Comparing x -coordinates, we have

$$\frac{9k + 3}{k + 1} = 5$$

$$\Rightarrow 9k + 3 = 5k + 5$$

$$\Rightarrow 4k = 2 \Rightarrow k = \frac{2}{4} = \frac{1}{2}$$

Comparing y -coordinates, we have

$$\frac{8k + 2}{k + 1} = 4$$

$$\Rightarrow k + 1 = 2k + 2$$

$$\Rightarrow k = \frac{1}{2}$$

Similarly, comparing z -coordinates we get $k = \frac{1}{2}$.

Thus, points P , Q and R are collinear.

Also, Q divides PR in the ratio $1 : 2$.

ILLUSTRATION 1.23

Consider two points $P(-2, 3, 5)$ and $Q(1, -4, 3)$. Find the point where PQ meets

- (i) yz -plane (ii) xz -plane and (iii) xy -plane

Sol. Let point R divides joint of $P(-2, 3, 5)$ and $Q(1, -4, 3)$ in the ratio $\lambda_1 : 1$.

Hence, by section formula, the coordinates of point of R_1 are

$$R \equiv \left(\frac{\lambda_1(1) - 2}{\lambda_1 + 1}, \frac{\lambda_1(-4) + 3}{\lambda_1 + 1}, \frac{\lambda_1(3) + 5}{\lambda_1 + 1} \right) \quad (1)$$

- (i) If point R lies on yz -plane then x -coordinate of R will be zero.

$$\therefore \frac{\lambda_1(1) - 2}{\lambda_1 + 1} = 0 \text{ or } \lambda_1 = 2$$

Putting this value of λ_1 in (1), we get

$$R \equiv \left(0, \frac{-5}{3}, \frac{11}{3} \right)$$

- (ii) If point R lies on xz -plane then y -coordinate of R will be zero.

$$\therefore \frac{\lambda_1(-4) + 3}{\lambda_1 + 1} = 0 \text{ or } \lambda_1 = \frac{3}{4}$$

Putting this value of λ_1 in (1), we get

$$R \equiv \left(\frac{-5}{7}, 0, \frac{29}{7} \right)$$

- (iii) If point R lies on xy -plane then z -coordinate of R will be zero.

$$\therefore \frac{\lambda_1(3) + 5}{\lambda_1 + 1} = 0 \text{ or } \lambda_1 = -\frac{5}{3}$$

Putting this value of λ_1 in (1), we get

$$R \equiv \left(\frac{11}{2}, -\frac{29}{2}, 0 \right)$$

ILLUSTRATION 1.24

Prove that line joining $A(1, -2, -3)$ and $B(-3, 4, 6)$ intersect x -axis. Also, find the point on the x -axis.

Sol. Let line joining A and B intersect x -axis at $C(x, 0, 0)$.

Let $AC : CB = k : 1$

$$\therefore C \equiv \left(\frac{-3k+1}{k+1}, \frac{4k-2}{k+1}, \frac{6k-3}{k+1} \right) \equiv (x, 0, 0)$$

$$4k - 2 = 0 \text{ and } 6k - 3 = 0$$

So, from both the equations $k = \frac{1}{2}$.

We would not have got the same value of k if line had not met x -axis. Thus, line AB meets x -axis.

$$\text{Also, } C \equiv \frac{-1}{3}, 0, 0$$

ILLUSTRATION 1.25

Prove that three diagonals of a cube intersect. Also find the acute angle between any two diagonals.

Sol. Cube with side length ' a ' and having three of its sides along positive coordinate axes is as shown in the figure.

Three diagonals are OP , BE and FC .

Midpoint of OP has coordinates $\left(\frac{a}{2}, \frac{a}{2}, \frac{a}{2}\right)$.

Also, midpoint of BE and FC has coordinates $\left(\frac{a}{2}, \frac{a}{2}, \frac{a}{2}\right)$.

Thus, diagonals are concurrent at $\left(\frac{a}{2}, \frac{a}{2}, \frac{a}{2}\right) \equiv M$.

Let us find angle between diagonals OP and AD .

Consider triangle MPD .

$$OP = \sqrt{a^2 + a^2 + a^2} = \sqrt{3}a$$

$$\therefore MP = MD = \frac{\sqrt{3}a}{2}$$

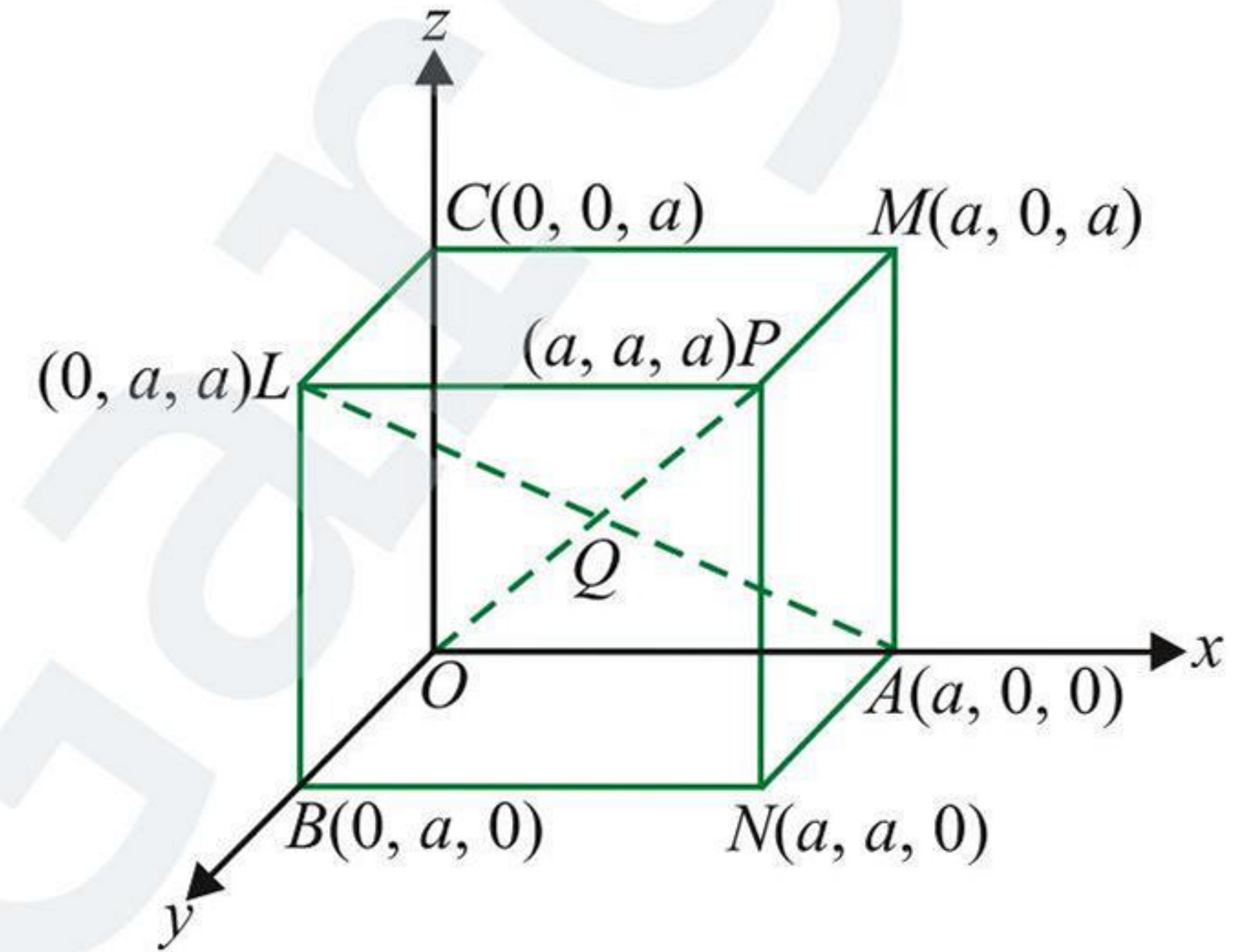
Also, $PD = \sqrt{2}a$

Now in triangle MPD , using cosine rule,

$$\begin{aligned} \cos \theta &= \frac{(MP)^2 + (MD)^2 - (PD)^2}{2 \times MP \times MD} = \frac{\frac{3}{4}a^2 + \frac{3}{4}a^2 - 2a^2}{2 \times \frac{\sqrt{3}a}{2} \times \frac{\sqrt{3}a}{2}} \\ &= \frac{-\frac{1}{2}a^2}{\frac{3a^2}{2}} = -\frac{1}{3} \end{aligned}$$

$$\therefore \theta = \cos^{-1}\left(-\frac{1}{3}\right) = \pi - \cos^{-1}\frac{1}{3}$$

Thus, acute angle between diagonals is $\cos^{-1}\frac{1}{3}$.

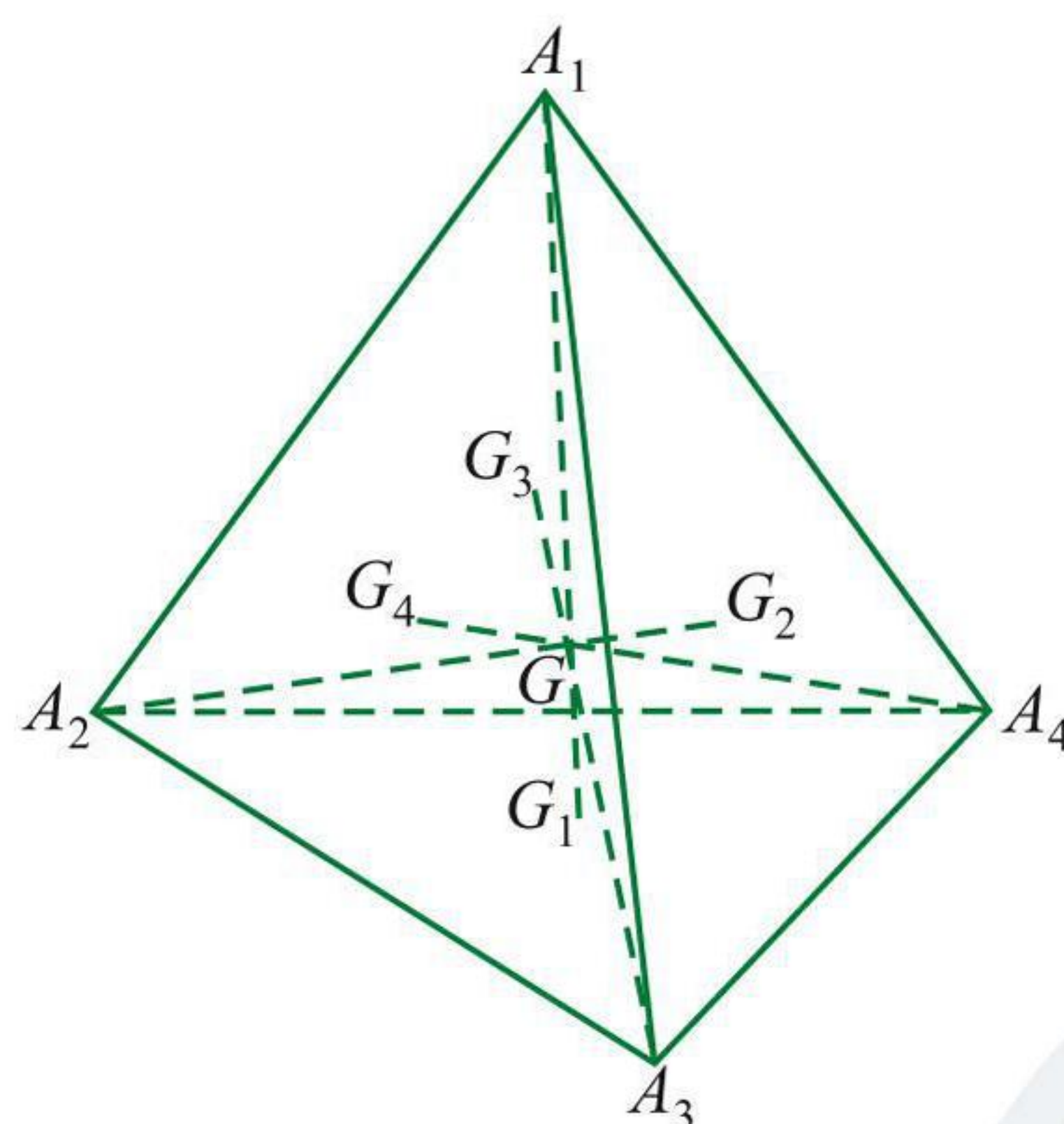
**ILLUSTRATION 1.26**

Prove that the line segments joining vertex to the centroid of the opposite face of tetrahedron are concurrent.

Sol. Consider tetrahedron $A_1A_2A_3A_4$ with $A_i \equiv (x_i, y_i, z_i)$; $i = 1, 2, 3, 4$.

Centroid of triangle $A_2A_3A_4$ is

$$G_1 \equiv \left(\frac{x_2 + x_3 + x_4}{3}, \frac{y_2 + y_3 + y_4}{3}, \frac{z_2 + z_3 + z_4}{3} \right)$$



Now, we have point G on A_1G_1 which divides it in the ratio $3 : 1$.

$$\therefore G \equiv \left(\frac{3\left(\frac{x_2+x_3+x_4}{3}\right) + 1(x_1)}{3+1}, \frac{3\left(\frac{y_2+y_3+y_4}{3}\right) + 1(y_1)}{3+1}, \frac{3\left(\frac{z_2+z_3+z_4}{3}\right) + 1(z_1)}{3+1} \right)$$

$$\equiv \left(\frac{x_1 + x_2 + x_3 + x_4}{4}, \frac{y_1 + y_2 + y_3 + y_4}{4}, \frac{z_1 + z_2 + z_3 + z_4}{4} \right)$$

Centroid of triangle $A_1A_3A_4$ is $G_2 \equiv \left(\frac{x_1 + x_3 + x_4}{3}, \frac{y_1 + y_3 + y_4}{3}, \frac{z_1 + z_3 + z_4}{3} \right)$.

Now, we have point on A_2G_2 which divides it in the ratio $3 : 1$, having coordinates

$$\left(\frac{x_1 + x_2 + x_3 + x_4}{4}, \frac{y_1 + y_2 + y_3 + y_4}{4}, \frac{z_1 + z_2 + z_3 + z_4}{4} \right) \equiv G$$

Thus, A_1G_1 and A_2G_2 intersect at G .

Similarly, A_3G_3 and A_4G_4 pass through point G , where G_3 and G_4 are centroids of triangular faces opposite to vertices A_3 and A_4 , respectively.

CONCEPT APPLICATION EXERCISE 1.3

1. Using section formula, prove that points $A(3, 2, -4)$, $B(5, 4, -6)$ and $C(9, 8, -10)$ are collinear.
2. Find the coordinates of the points which trisect the line segment joining the points $P(4, 2, -6)$ and $Q(10, -16, 6)$.
3. Equation $ax + by + cz + d = 0$ represents the equation of plane in the space. Prove that points $A(2, 3, -4)$ and $B(7, 2, 0)$ lie on the same side of the plane $x - y + 2z - 7 = 0$.
4. Prove that line segments joining midpoints of opposite edges of tetrahedron are concurrent.

ANSWERS

2. $(8, -10, 2)$

SPHERE

In 3D coordinate system, a sphere is a locus of variable point $P(x, y, z)$ whose distance from fixed point $C(x_0, y_0, z_0)$ is constant r . The fixed point is called centre and constant distance r is called radius of the sphere.

$$\text{Thus, } CP = r \text{ or } CP^2 = r^2$$

So, $(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = r^2$, which is equation of the sphere.

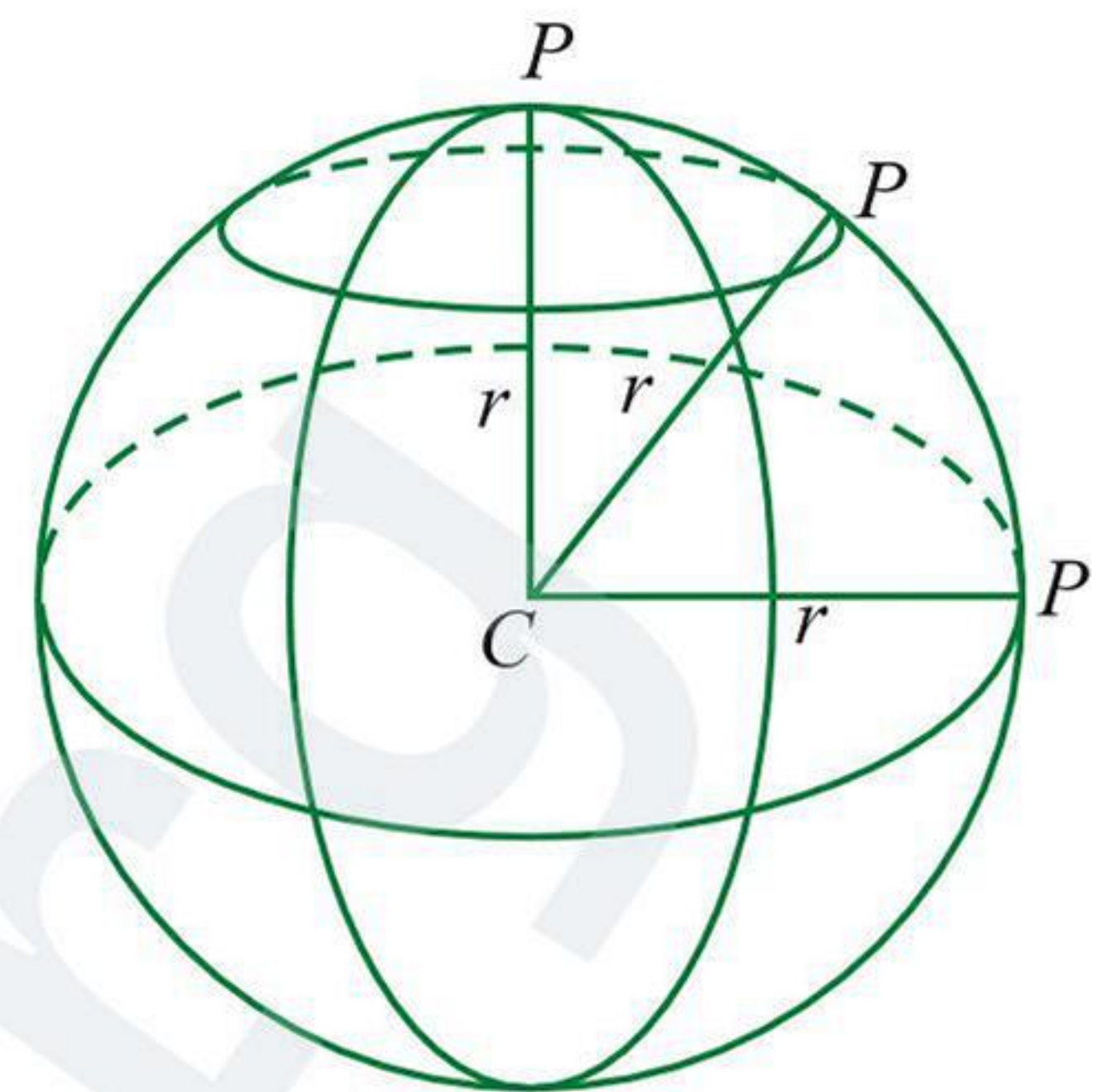


ILLUSTRATION 1.27

Find the equation of a sphere with centre $(2, -1, 3)$ and radius 4.

Sol. The equation of sphere is given by

$$(x - 2)^2 + (y - (-1))^2 + (z - 3)^2 = 4^2,$$

which simplifies to $x^2 + y^2 + z^2 - 4x + 2y - 6z = 2$

ILLUSTRATION 1.28

Find the equation of a sphere on which the two points $A(4, 1, -1)$ and $B(6, 7, 9)$ lie directly opposite one another (that is, the line through them forms a diameter of the sphere).

Sol. Clearly the centre is the midpoint of A and B , which is

$$C \equiv \left(\frac{4+6}{2}, \frac{1+7}{2}, \frac{-1+9}{2} \right) \equiv (5, 4, 4).$$

Radius of the sphere is

$$\begin{aligned} CA &= \sqrt{(5-4)^2 + (4-1)^2 + (4-(-1))^2} \\ &= \sqrt{1^2 + 3^2 + 5^2} = \sqrt{35} \end{aligned}$$

$\therefore$ Equation of sphere is $(x - 5)^2 + (y - 4)^2 + (z - 4)^2 = 35$, which simplifies to

$$x^2 + y^2 + z^2 - 10x - 8y - 8z + 22 = 0$$

ILLUSTRATION 1.29

Find the equation of the largest possible sphere with centre $(3, 2, 4)$ that is fully contained within the first octant. (tangentially “touching” a coordinate plane is permissible).

Sol. Clearly the largest possible sphere is that which is touching all the coordinate planes.

Since the y -coordinate of 2 is the smallest of the three coordinates and is 2 units from the xz -coordinate plane, this will be the radius.

Thus, equation of the sphere will be $(x - 3)^2 + (y - 2)^2 + (z - 4)^2 = 4$, which simplifies to

$$x^2 + y^2 + z^2 - 6x - 4y - 8z + 21 = 0$$

ILLUSTRATION 1.30

The sphere $(x + 6)^2 + (y - 1)^2 + (z - 4)^2 = 100$ intersects the yz -coordinate plane, forming a circle. What is the radius of this circle?

Sol. The points where sphere intersect yz -plane, $x = 0$.

So, putting $x = 0$ in the given equation of sphere we have

$$((0) + 6)^2 + (y - 1)^2 + (z - 4)^2 = 100$$

or
$$(y - 1)^2 + (z - 4)^2 = 64$$

Clearly this is equation of circle in yz -plane.

The centre of the circle is $(1, 4)$ and radius is '8'.

GENERAL EQUATION OF SPHERE

From the previous discussion we get that equation of sphere takes the form

$x^2 + y^2 + z^2 + 2gx + 2fy + 2hz + c = 0$, which can put in centre-radius form as

$$(x + g)^2 + (y + f)^2 + (z + h)^2 = g^2 + f^2 + h^2 - c$$

Centre of the sphere is $(-g, -f, -h)$ and radius is $\sqrt{g^2 + f^2 + h^2 - c}$

ILLUSTRATION 1.31

Find the centre and radius of the sphere $x^2 + y^2 + z^2 - 2x + 6y - 4z = 22$.

Sol. The given equation can be put in the form

$$(x - 1)^2 + (y + 3)^2 + (z - 2)^2 = 36$$

Thus, the sphere has a centre of $(-1, 3, -2)$ and a radius of $r = 6$.

ILLUSTRATION 1.32

Find the intercept of sphere $(x + 1)^2 + (y - 4)^2 + (z - 6)^2 = 33$ on z -axis.

Sol. When sphere cuts x -axis, $x = 0$ and $y = 0$.

So, from the equation of sphere, we have

$$((0) + 1)^2 + ((0) - 4)^2 + (z - 6)^2 = 33$$

$$\Rightarrow 1^2 + (-4)^2 + (z - 6)^2 = 33$$

$$\Rightarrow (z - 6)^2 = 16$$

$$\Rightarrow z - 6 = \pm 4$$

$$\Rightarrow z = 10 \text{ or } 2$$

Thus sphere intersect z -axis at $A(0, 0, 2)$ and $B(0, 0, 10)$.

So, $AB = 8$

ILLUSTRATION 1.33

Find the equation of the sphere which passes through $(1, 0, 0)$ $(0, 1, 0)$, $(0, 0, 1)$ and has radius as small as possible.

Sol. Let the equation of the required sphere be

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0 \quad (1)$$

As (1) passes through (1, 0, 0), (0, 1, 0) and (0, 0, 1), we get $1 + 2u + d = 0$, $1 + 2v + d = 0$, $1 + 2w + d = 0$

$$\Rightarrow u = v = w = -\frac{1}{2}(d + 1)$$

If R is the radius of (1), then $R^2 = u^2 + v^2 + w^2 - d$

$$\begin{aligned} \Rightarrow R^2 &= \frac{3}{4}(d + 1)^2 - d \\ &= \frac{3}{4}\left[d^2 + 2d + 1 - \frac{4}{3}d\right] \\ &= \frac{3}{4}\left[d^2 + \frac{2}{3}d + 1\right] \\ &= \frac{3}{4}\left[(d + 1/3)^2 + 1 - 1/9\right] = \frac{3}{4}\left[(d + 1/3)^2 + 8/9\right] \end{aligned}$$

The last equation shows that R^2 (and thus R) will be least if and only if $d = -1/3$.

$$\text{Therefore, } u = v = w = -\frac{1}{2}\left(1 - \frac{1}{3}\right) = -\frac{1}{3}$$

$$\text{Hence, the equation of the required sphere is } x^2 + y^2 + z^2 - \frac{2}{3}(x + y + z) - \frac{1}{3} = 0$$

$$\text{or } 3(x^2 + y^2 + z^2) - 2(x + y + z) - 1 = 0$$

ILLUSTRATION 1.34

A sphere of constant radius k passes through the origin and meet the axes in A , B and C . Prove that the centroid of the triangle ABC lies on the sphere $9(x^2 + y^2 + z^2) = 4k^2$.

Sol. Let equation of any sphere passing through the origin and having radius k be

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz = 0 \quad (1)$$

As the radius of the sphere is k , we get

$$u^2 + v^2 + w^2 = k^2 \quad (2)$$

Note that (1) meets x -axis in $O(0, 0, 0)$ and $A(-2u, 0, 0)$; y -axis in $O(0, 0, 0)$ and $B(0, -2v, 0)$ and z -axis in $O(0, 0, 0)$ and $C(0, 0, -2w)$.

Suppose the centroid of the triangle ABC be (α, β, γ) .

$$\text{Then } \alpha = -\frac{2u}{3}, \beta = -\frac{2v}{3}, \gamma = -\frac{2w}{3}$$

$$\Rightarrow u = -\frac{3\alpha}{2}, v = -\frac{3\beta}{2}, w = -\frac{3\gamma}{2}$$

$$\text{Putting this in (2), we get } \left(-\frac{3}{2}\alpha\right)^2 + \left(-\frac{3}{2}\beta\right)^2 + \left(-\frac{3}{2}\gamma\right)^2 = k^2$$

$$\Rightarrow \alpha^2 + \beta^2 + \gamma^2 = \frac{4}{9}k^2.$$

This show that centroid of the triangle ABC lies on $x^2 + y^2 + z^2 = \frac{4}{9}k^2$.

CONCEPT APPLICATION EXERCISE 1.4

1. Find the equation of the sphere whose centre is $(3, 1, 2)$ and radius is 5.
2. Find the equation of the sphere passing through $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$.
3. Find the locus of a point which moves such that the sum of the squares of its distance from the points $A(1, 2, 3)$, $B(2, -3, 5)$ and $C(0, 7, 4)$ is 120.
4. A point $P(x, y, z)$ is such that $3PA = 2PB$ where A and B are the points $(1, 3, 4)$ and $(1, -2, -1)$ respectively. Find the equation to the locus of the point P and verify that the locus is a sphere. Find its centre and radius.

ANSWERS

1. $x^2 + y^2 + z^2 - 6x - 2y - 4z - 11 = 0$
2. $x^2 + y^2 + z^2 - x - y - z = 0$
3. $x^2 + y^2 + z^2 - 2x - 4y - 8z - 1 = 0$
4. $x^2 + y^2 + z^2 - 2x - 14y - 16z + 42 = 0$; Centre $\equiv (1, 7, 8)$; Radius $= 6\sqrt{2}$

Exercises

Single Correct Answer Type

- The distance of point $P(3, 4, 5)$ from the xy -plane is
 (1) 3 units (2) 4 units (3) 5 units (4) 550
- The length of perpendicular drawn from the point $P(-3, -4, 5)$ on y -axis
 (1) $\sqrt{41}$ (2) $\sqrt{34}$ (3) 5 (4) none of these
- M is the foot of the perpendicular drawn from a point $P(-1, 7, -5)$ on the xy -plane. The coordinates of point M are
 (1) $(-1, 7, 0)$ (2) $(1, -7, 0)$ (3) $(7, 5, 0)$ (4) $(-1, 6, 0)$
- If feet of perpendiculars from point $P(1, 2, 3)$ on x , y and z -axis are A , B and C respectively, then distance of centroid of triangle ABC from origin is
 (1) $\frac{\sqrt{14}}{3}$ (2) $\frac{\sqrt{14}}{2}$ (3) $\sqrt{14}$ (4) $3\sqrt{14}$
- If feet of perpendiculars from point $P(1, 2, 3)$ on xy , yz and zx -plane are A , B and C respectively, then distance of centroid of triangle ABC from origin is
 (1) $\frac{\sqrt{14}}{3}$ (2) $\frac{2\sqrt{14}}{3}$ (3) $\frac{3\sqrt{14}}{2}$ (4) $3\sqrt{14}$
- A point R with x -coordinate 6 lies on the line segment joining the points $P(2, -3, 4)$ and $Q(8, 0, 10)$. Then distance of R from origin is
 (1) $\sqrt{47}$ (2) $\sqrt{41}$ (3) $\sqrt{53}$ (4) $3\sqrt{7}$
- The volume of the cuboid having combined equations of opposite faces as $x^2 - 3x - 4 = 0$, $y^2 - 5y + 6 = 0$ and $z^2 + 6z + 8 = 0$ is
 (1) 20 cu. units (2) 10 cu. units (3) 15 cu. units (4) 24 cu. units
- Number of spheres touching all the coordinate planes and having fixed radius is
 (1) 2 (2) 4 (3) 6 (4) 8
- If the sum of the squares of the distance of a point from the three coordinate axes be 36, then its distance from the origin is
 (1) 5 (2) $3\sqrt{2}$ (3) $\sqrt{21}$ (4) $3\sqrt{3}$
- The xy -plane divides the line joining the points $(-1, 3, 4)$ and $(2, -5, 6)$
 (1) internally in the ratio 2 : 3 (2) externally in the ratio 2 : 3
 (3) internally in the ratio 3 : 2 (4) externally in the ratio 4 : 3
- If the points $(-1, 3, 2)$, $(-4, 2, -2)$ and $(5, 5, \lambda)$ are collinear, then $\lambda =$
 (1) 10 (2) -10 (3) -8 (4) 8
- From which of the following the distance of the point $(1, 2, 3)$ is $\sqrt{10}$?
 (1) origin (2) x -axis (3) y -axis (4) z -axis
- Distance of the point $(1, 2, 3)$ from the coordinate axes are
 (1) 13, 10, 5 (2) $\frac{1}{\sqrt{13}}, \frac{1}{\sqrt{10}}, \frac{1}{\sqrt{5}}$ (3) $\sqrt{5}, \sqrt{13}, \sqrt{10}$ (4) $\sqrt{13}, \sqrt{10}, \sqrt{5}$

14. If the centroid of triangle whose vertices are $(a, 1, 3)$, $(-2, b, -5)$ and $(4, 7, c)$ be the origin, then the value of a, b and c are
 (1) $-2, -8, -2$ (2) $2, 8, -2$ (3) $-2, -8, 2$ (4) $7, -1, 0$
15. The point of intersection of the line joining the points $(3, 4, 1)$ and $(5, 1, 6)$ and the xy -plane is
 (1) $(13, 23, 0)$ (2) $\left(\frac{13}{5}, \frac{23}{5}, 0\right)$ (3) $(-13, 23, 0)$ (4) $\left(-\frac{13}{5}, \frac{23}{5}, 0\right)$
16. The coordinates of a point which is equidistance from the points $(0, 0, 0)$, $(a, 0, 0)$, $(0, b, 0)$ and $(0, 0, c)$ are given by
 (1) $\left(\frac{a}{2}, \frac{b}{2}, \frac{c}{2}\right)$ (2) $\left(\frac{a}{3}, \frac{b}{3}, \frac{c}{3}\right)$ (3) $\left(\frac{a}{4}, \frac{b}{4}, \frac{c}{4}\right)$ (4) $\left(-\frac{a}{2}, -\frac{b}{2}, -\frac{c}{2}\right)$
17. The coordinates of a point P are $(3, 12, 4)$ with respect to origin O , then the direction cosines of OP are
 (1) $3, 12, 4$ (2) $\frac{1}{4}, \frac{1}{3}, \frac{1}{2}$ (3) $\frac{3}{\sqrt{13}}, \frac{1}{\sqrt{13}}, \frac{2}{\sqrt{13}}$ (4) $\frac{3}{13}, \frac{12}{13}, \frac{4}{13}$
18. A line joining point P with the origin makes angles of 45° and 60° with the positive axes of X and Y respectively. The angle made by the same line with the positive axis of Z , is
 (1) 30° or 60° (2) 60° or 90° (3) 90° or 120° (4) 60° or 120°
19. The points $A(5, -1, 1)$, $B(7, -4, 7)$, $C(1, -6, 10)$ and $D(-1, -3, 4)$ are vertices of a
 (1) square (2) rhombus (3) rectangle (4) None of these
20. Which of the following point lies inside the sphere $x^2 + y^2 - 4x - 6y + 8z - 7 = 0$?
 (1) $(1, -2, 1)$ (2) $(0, 0, 3)$ (3) $(0, 1, 2)$ (4) $(1, 2, 0)$
21. Number of points common to the spheres $x^2 + y^2 + z^2 = 36$ and $x^2 + y^2 + z^2 - 20x + 84 = 0$ is
 (1) zero (2) one (3) infinite (4) none of these
22. The xy -plane cuts the sphere $x^2 + y^2 + z^2 - 2x - 4y + 6z - 4 = 0$ in a circle, the radius of which is
 (1) 2 (2) 3 (3) 4 (4) 5
23. If equation $6x + 3y - 2z = 18$ represents the equation of plane in the space then the centroid of triangle ABC where A, B, C are the points of intersection of the planes with the coordinate axes is
 (1) $(1, 2, -3)$ (2) $(-1, 2, 3)$ (3) $(-1, -2, -3)$ (4) $(1, -2, 3)$

Answers Key

EXERCISES

Single Correct Answer Type

1. (3) 2. (2) 3. (1) 4. (1) 5. (2) 6. (3) 7. (2) 8. (4) 9. (2) 10. (2)
 11. (1) 12. (3) 13. (4) 14. (3) 15. (2) 16. (1) 17. (4) 18. (4) 19. (2) 20. (4)
 21. (2) 22. (2) 23. (1)

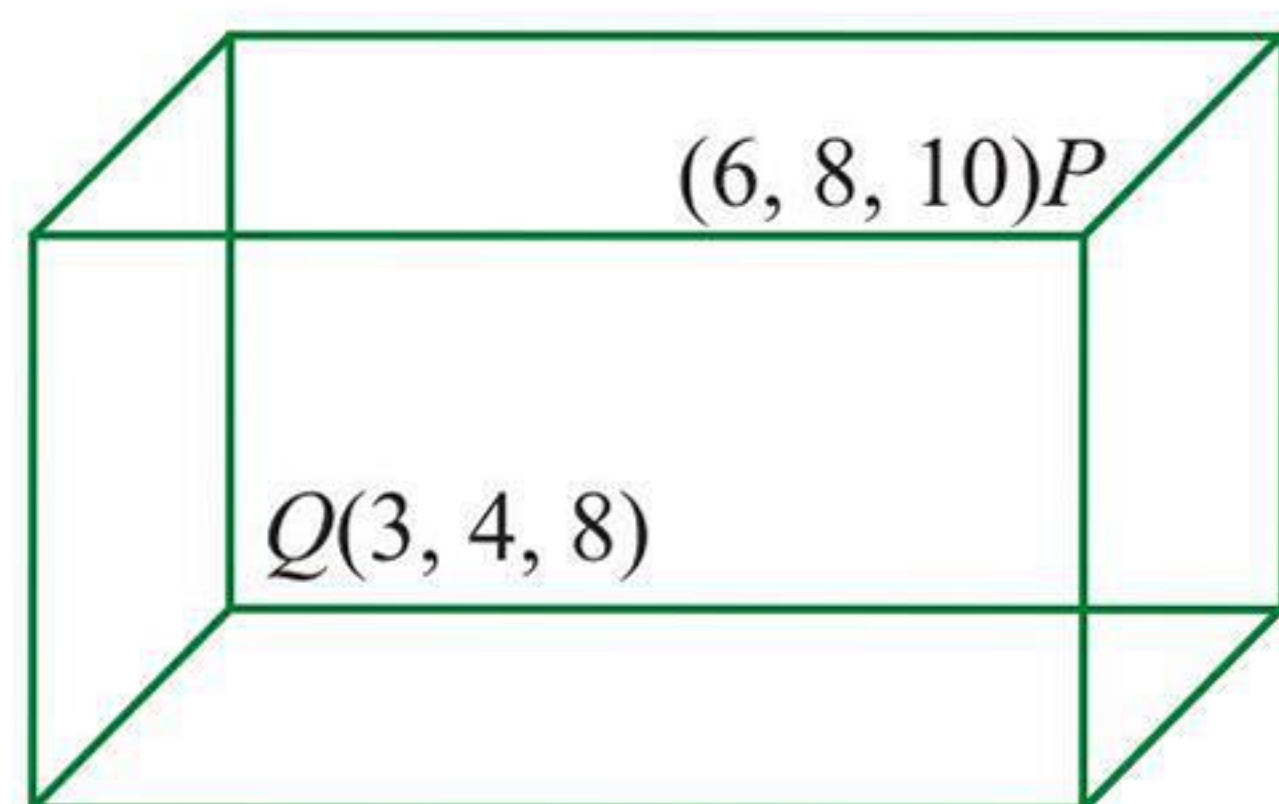
Solutions

Chapter 1

Concept Application Exercises

Exercise 1.1

1.



From the figure it is clear that the length of the edges of the parallelepiped a, b, c are $x_2 - x_1, y_2 - y_1, z_2 - z_1$ or $6 - 3, 8 - 4, 10 - 8$ or $3, 4, 2$.

Therefore, the length of the diagonal will be

$$\sqrt{a^2 + b^2 + c^2} = \sqrt{9 + 16 + 4} = \sqrt{39}.$$

2. We have

$$x^2 - 2x - 3 = 0 \quad (1)$$

$$\Rightarrow (x - 3)(x + 1) = 0$$

$$\Rightarrow x = 3 \text{ and } x = -1$$

Thus, equation (1) represents combined equation of two planes $x = -1$ and $x = 3$ which are parallel to yz -plane.

Distance between two planes is 4 units.

3. (i) Distance of plane $z = 6$ from xy -plane is 6 units.

Distance of point $P(3, 4, -5)$ from xy -plane is 5 units.

But point P and plane $z = 6$ lie on the opposite sides of xy -plane.

So, distance of point P from plane $z = 6$ is $5 + 6 = 11$ units.

(ii) Distance of plane $x = 7$ from yz -plane is 7 units.

Distance of point $P(3, 4, -5)$ from yz -plane is 3 units.

Since point P and plane $x = 7$ lie on the same side of yz -plane, required distance is $7 - 3 = 4$ units.

4. Point $P(-3, 2, 4)$ lies in second octant.

Image of P in xy -plane is $P'(-3, 2, -4)$ which lies in sixth octant.

Image of P in yz -plane is $P''(3, 2, 4)$ which lies in first octant.

Image of P in xz -plane is $P'''(-3, -2, 4)$ which lies in third octant.

5. Here, $\cos^2 \alpha + \cos^2 (90^\circ - \alpha) + \cos^2 \gamma = 1$

$$\Rightarrow \cos^2 \alpha + \sin^2 \alpha + \cos^2 \gamma = 1$$

$$\Rightarrow \cos^2 \gamma + 1 = 1 \Rightarrow \gamma = 90^\circ$$

6. $\cos 2\alpha + \cos 2\beta + \cos 2\gamma$

$$= 2 \cos^2 \alpha - 1 + 2 \cos^2 \beta - 1 + 2 \cos^2 \gamma - 1$$

$$= 2(\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma) - 3 = 2 - 3 = -1$$

7. We have point $P(2, 3, 6)$.

$$\therefore \cos \alpha = \frac{2}{\sqrt{2^2 + 3^2 + 6^2}} = \frac{2}{7},$$

$$\cos \beta = \frac{3}{7} \text{ and } \cos \gamma = \frac{6}{7}$$

Hence, angle α is greatest as cosine is decreasing function for angles lying in $(0, \pi)$.

Exercise 1.2

1. (i) Let points $(0, 7, -10)$, $(1, 6, -6)$, and $(4, 9, -6)$ be denoted by A, B , and C , respectively.

$$\begin{aligned} AB &= \sqrt{(1-0)^2 + (6-7)^2 + (-6+10)^2} \\ &= \sqrt{1+1+16} = \sqrt{18} = 3\sqrt{2} \end{aligned}$$

$$\begin{aligned} BC &= \sqrt{(4-1)^2 + (9-6)^2 + (-6+6)^2} \\ &= \sqrt{3^2+3^2} = \sqrt{18} = 3\sqrt{2} \end{aligned}$$

$$\begin{aligned} CA &= \sqrt{(0-4)^2 + (7-9)^2 + (-10+6)^2} \\ &= \sqrt{(-4)^2 + (-2)^2 + (-4)^2} = \sqrt{36} = 6 \end{aligned}$$

Here, $AB = BC \neq CA$

Thus, the given points are the vertices of an isosceles triangle.

- (ii) Let $(-1, 2, 1)$, $(1, -2, 5)$, $(4, -7, 8)$, and $(2, -3, 4)$ be denoted by A , B , C , and D , respectively.

$$AB = \sqrt{(1+1)^2 + (-2-2)^2 + (5-1)^2}$$

$$= \sqrt{4+16+16} = \sqrt{36} = 6$$

$$BC = \sqrt{(4-1)^2 + (-7+2)^2 + (8-5)^2}$$

$$= \sqrt{9+25+9} = \sqrt{43}$$

$$CD = \sqrt{(2-4)^2 + (-3+7)^2 + (4-8)^2}$$

$$= \sqrt{4+16+16} = \sqrt{36} = 6$$

$$DA = \sqrt{(-1-2)^2 + (2+3)^2 + (1-4)^2}$$

$$= \sqrt{9+25+9} = \sqrt{43}$$

Here, $AB = CD = 6$, $BC = AD = \sqrt{43}$

Hence, the opposite sides of quadrilateral $ABCD$, whose vertices are taken in order, are equal.

Therefore, $ABCD$ is a parallelogram.

Hence, the given points are the vertices of a parallelogram.

2. Let point P be (x, y, z)

$$\text{Now } x^2 + y^2 + z^2 = 49 \quad (1)$$

$$y^2 + z^2 = 13 \quad (2)$$

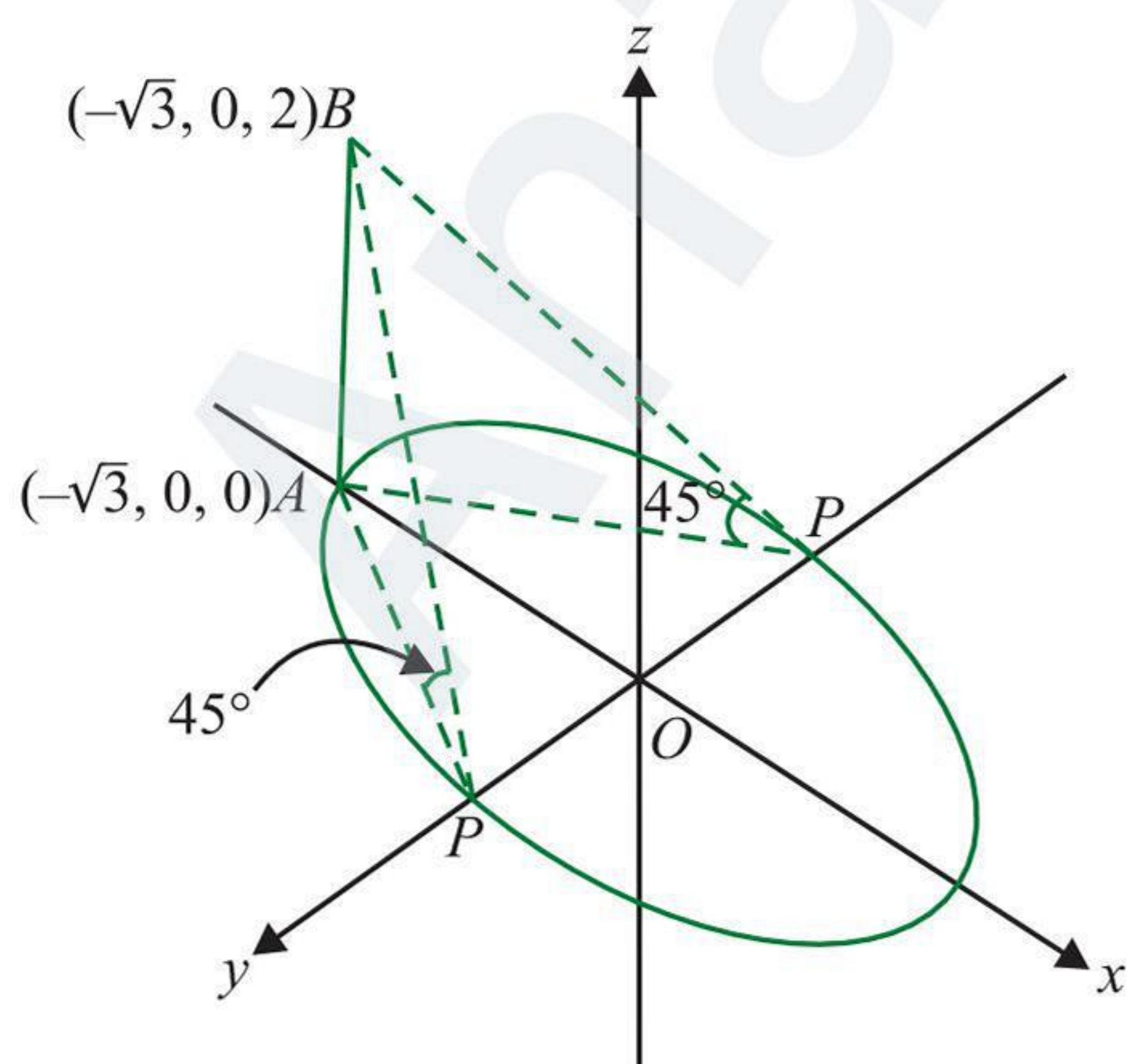
$$\text{and } x^2 + z^2 = 40 \quad (3)$$

$$\therefore x^2 = 36, y^2 = 9 \text{ and } z^2 = 4$$

$$\therefore (x, y, z) \equiv (\pm 6, \pm 3, \pm 2)$$

Thus, there are eight points.

3. Clearly, ellipse $x^2 + 3y^2 = 3$ lies in xy -plane.



Point P lies on

$$x^2 + 3y^2 = 3 \quad (1)$$

Now, from diagram according to given conditions $AP = AB$

$$\text{or } (x + \sqrt{3})^2 + (y - 0)^2 = 4$$

$$\text{or } (x + \sqrt{3})^2 + y^2 = 4 \quad (2)$$

Solving (1) and (2), we get $x = 0$ and $y = \pm 1$.

So, point P has coordinates $(0, \pm 1, 0)$.

4. We have $P(3, y, z)$. This point lies on the plane $x = 3$, which is parallel to yz -plane.

Given that $OP = 5$, where O is origin.

$$\therefore 9 + y^2 + z^2 = 25$$

$$\text{or } y^2 + z^2 = 16.$$

This is equation of circle of radius 4 units and centre at $(3, 0, 0)$ lying in the plane $x = 3$.

Exercise 1.3

1. Let C divide AB in the ratio $\lambda : 1$. Then

$$C \equiv \left(\frac{5\lambda + 3}{\lambda + 1}, \frac{4\lambda + 2}{\lambda + 1}, \frac{-6\lambda - 4}{\lambda + 1} \right) \equiv (9, 8 - 10)$$

Comparing x -coordinate, we get

$$5\lambda + 3 = 9\lambda + 9 \text{ or } 4\lambda = -6$$

$$\therefore \lambda = -3/2$$

Also, from $4\lambda + 2 = 8\lambda + 8$ and $-6\lambda - 4 = -10\lambda - 10$, we get the same value of λ .

Thus, points are collinear.

2. Let A and B be the points that trisect the line segment joining points $P(4, 2, -6)$ and $Q(10, -16, 6)$.



Point A divides PQ in the ratio $1 : 2$. Therefore, by section formula, the coordinates of point A are given by

$$\left(\frac{1(10) + 2(4)}{1 + 2}, \frac{1(-16) + 2(2)}{1 + 2}, \frac{1(6) + 2(-6)}{1 + 2} \right)$$

$$= (6, -4, -2)$$

Point B divides PQ in the ratio $2 : 1$. Therefore, by section formula, the coordinates of point B are given by

$$\left(\frac{2(10) + 1(4)}{2 + 1}, \frac{2(-16) + 1(2)}{2 + 1}, \frac{2(6) + 1(-6)}{2 + 1} \right)$$

$$= (8, -10, 2)$$

Thus, $(6, -4, -2)$ and $(8, -10, 2)$ are the points that trisect the line segment joining points $P(4, 2, -6)$ and $Q(10, -16, 6)$.

3. Let given plane divides the join of A and B at point P in the ratio $\lambda : 1$.

$$\therefore P \equiv \left(\frac{7\lambda+2}{\lambda+1}, \frac{2\lambda+3}{\lambda+1}, \frac{-4}{\lambda+1} \right)$$

Point P lies on the given plane.

$$\text{So, } \frac{7\lambda+2}{\lambda+1} - \frac{2\lambda+3}{\lambda+1} - \frac{8}{\lambda+1} - 7 = 0$$

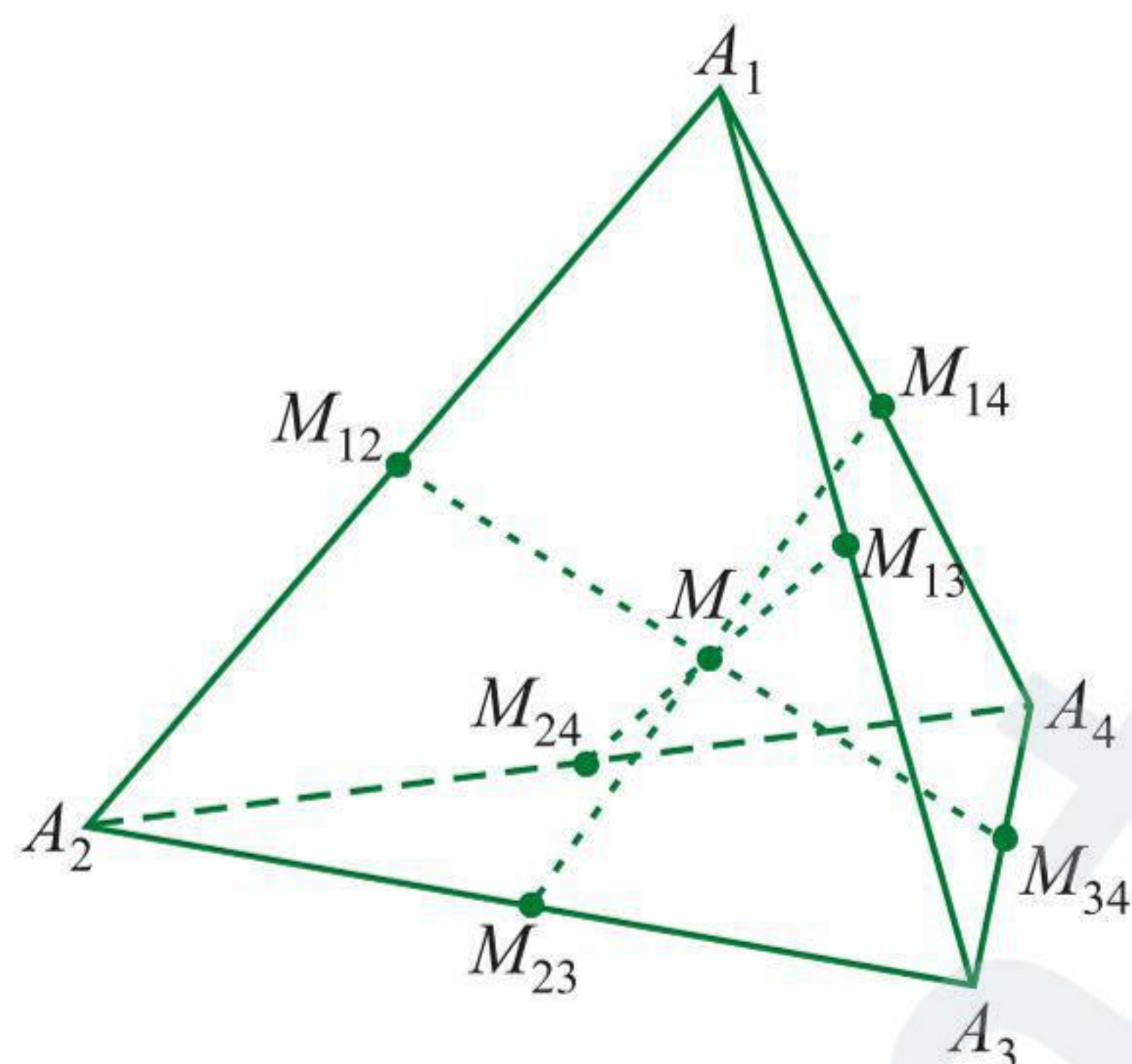
$$\text{or } 5\lambda - 9 - 7\lambda - 7 = 0$$

$$\text{or } \lambda = -8$$

Thus point P divides AB externally in the ratio $8 : 1$.

So, points A and B lies on the same side of the plane.

4. Consider tetrahedron $A_1A_2A_3A_4$ with $A_i \equiv (x_i, y_i, z_i)$; $i = 1, 2, 3, 4$.



We have three pairs of opposite edges, (A_1A_2, A_3A_4) , (A_1A_3, A_2A_4) and (A_1A_4, A_2A_3) .

Midpoint of A_1A_2 is

$$M_{12} \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2} \right).$$

Midpoint of A_3A_4 is

$$M_{34} \left(\frac{x_3+x_4}{2}, \frac{y_3+y_4}{2}, \frac{z_3+z_4}{2} \right).$$

Now, midpoint of $M_{12}M_{34}$ is

$$M \left(\frac{\frac{x_1+x_2}{2} + \frac{x_3+x_4}{2}}{2}, \frac{\frac{y_1+y_2}{2} + \frac{y_3+y_4}{2}}{2}, \frac{\frac{z_1+z_2}{2} + \frac{z_3+z_4}{2}}{2} \right)$$

$$\equiv M \left(\frac{\sum_{i=1}^4 x_i}{4}, \frac{\sum_{i=1}^4 y_i}{4}, \frac{\sum_{i=1}^4 z_i}{4} \right)$$

We can show that point M also bisect the line segment joining midpoints of other two pairs of opposite edges.

Exercise 1.4

1. Equation of the sphere whose centre is $(3, 1, 2)$ and radius is 5 is

$$(x-3)^2 + (y-1)^2 + (z-2)^2 = 5^2$$

$$\text{or } x^2 + y^2 + z^2 - 6x - 2y - 4z - 11 = 0$$

2. Let the equation of the sphere passing through $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$ be

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0 \quad (1)$$

As (1) passes through $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$ we must have $d = 0$; $1 + 2u + d = 0$; $1 + 2v + d = 0$; $1 + 2w + d = 0$

Since $d = 0$, we get $2u = 2v = 2w = -1$.

Thus, the equation of the required sphere is $x^2 + y^2 + z^2 - x - y - z = 0$.

3. Let $P(x, y, z)$ be any point on the locus. Then

$$PA^2 + PB^2 + PC^2 = 120$$

$$\Rightarrow (x-1)^2 + (y-2)^2 + (z-3)^2 + (x-2)^2 + (y+3)^2 + (z-5)^2 + (x-0)^2 + (y-7)^2 + (z-4)^2 = 120$$

$$\text{or } 3x^2 + 3y^2 + 3z^2 - 6x - 12y - 24z + 117 = 120$$

$$\text{or } x^2 + y^2 + z^2 - 2x - 4y - 8z - 1 = 0$$

This represents a sphere with centre at $(1, 2, 4)$ and radius as $\sqrt{1^2 + 2^2 + 4^2 + 1} = \sqrt{22}$.

4. Since $3PA = 2PB$, we get

$$9PA^2 = 4PB^2$$

$$\Rightarrow 9[(x-1)^2 + (y-3)^2 + (z-4)^2] = 4[(x-1)^2 + (y+2)^2 + (z+1)^2]$$

$$\Rightarrow 9[x^2 + y^2 + z^2 - 2x - 6y - 8z + 26] = 4[x^2 + y^2 + z^2 - 2x + 4y + 2z + 6]$$

$$\Rightarrow 5x^2 + 5y^2 - 10x - 70y - 80z + 210 = 0$$

$$\Rightarrow x^2 + y^2 + z^2 - 2x - 14y - 16z + 42 = 0$$

This represents a sphere with centre at $(1, 7, 8)$ and radius equal to $\sqrt{1^2 + 7^2 + 8^2 - 42} = \sqrt{72} = 6\sqrt{2}$.

Exercises

Single Correct Answer Type

1. (3) Given, point is $P(3, 4, 5)$.

Distance of P from xy -plane

$$= |z \text{ coordinate of } P| = 5$$

2. (2) We know that, on the y -axis, $x = 0$ and $z = 0$

Thus, foot of perpendicular from P on y -axis is $A(0, 4, 0)$.

$$\begin{aligned}\therefore PA &= \sqrt{(0-3)^2 + (4-4)^2 + (0-5)^2} \\ &= \sqrt{9+0+25} = \sqrt{34}\end{aligned}$$

3. (1) We know that, on the xy -plane $z = 0$.

Hence, the coordinates of the points L are $(-1, 7, 0)$.

4. (1) Clearly, $A \equiv (1, 0, 0)$, $B \equiv (0, 2, 0)$ and $C \equiv (0, 0, 3)$.

$\therefore$ Centroid of triangle ABC is

$$\begin{aligned}\therefore G &\equiv \left(\frac{1+0+0}{3}, \frac{0+2+0}{3}, \frac{0+0+3}{3} \right) \\ &\equiv \left(\frac{1}{3}, \frac{2}{3}, 1 \right)\end{aligned}$$

$$OG \equiv \sqrt{\left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2 + (1)^2} = \frac{\sqrt{14}}{3}$$

5. (2) Clearly $A \equiv (1, 2, 0)$, $B \equiv (0, 2, 3)$ and $C \equiv (1, 0, 3)$.

$\therefore$ Centroid of triangle ABC is

$$\begin{aligned}G &\equiv \left(\frac{1+0+1}{3}, \frac{2+2+0}{3}, \frac{0+3+3}{3} \right) \\ &\equiv \left(\frac{2}{3}, \frac{4}{3}, 2 \right)\end{aligned}$$

$$\therefore OG = \sqrt{\left(\frac{2}{3}\right)^2 + \left(\frac{4}{3}\right)^2 + (2)^2} = \frac{2\sqrt{14}}{3}$$

6. (3) Suppose R divides PQ in the ratio $k : 1$.

$$\therefore R \equiv \left(\frac{8k+2}{k+1}, \frac{-3}{k+1}, \frac{10k+4}{k+1} \right)$$

It is given that the x -coordinate of point R is 6.

$$\therefore \frac{8k+2}{k+1} = 6 \Rightarrow 8k+2 = 6k+6$$

$$\Rightarrow 2k = 4 \Rightarrow k = 2$$

Therefore, the coordinates of point R are given by

$$\left(6, \frac{-3}{2+1}, \frac{10 \times 2 + 4}{2+4} \right) \equiv (6, -1, 4)$$

7. (2) We have $x^2 - 3x - 4 = 0$

$$\Rightarrow (x-4)(x+1) = 0$$

$$\Rightarrow x = 4, x = -1$$

Distance between two parallel planes (parallel to yz -plane) is 5 units.

$$\text{We have } y^2 - 5y + 6 = 0$$

$$\Rightarrow (y-2)(y-3) = 0$$

$$\Rightarrow y = 2, y = 3$$

Distance between two parallel planes (parallel to xz -plane) is 1 unit.

$$\text{We have } z^2 + 6z + 8 = 0$$

$$\Rightarrow (z+2)(z+4) = 0$$

$$\Rightarrow z = -2, z = -4$$

Distance between two parallel planes (parallel to xy -plane) is 2 unit.

So, volume of cuboid is $5 \times 1 \times 2 = 10$ cu. units.

8. (4) Clearly, there will be eight spheres, one in each octant.

9. (2) Let $P(x, y, z)$. Now under given condition, we get

$$\left[\sqrt{x^2 + y^2} \right]^2 + \left[\sqrt{y^2 + z^2} \right]^2 + \left[\sqrt{z^2 + x^2} \right]^2 = 36$$

$$\Rightarrow x^2 + y^2 + z^2 = 18$$

Then distance from origin to the point (x, y, z) is

$$\sqrt{x^2 + y^2 + z^2} = \sqrt{18} = 3\sqrt{2}$$

10. (2) Let xy -plane divides the line joining the points $A(-1, 3, 4)$ and $B(2, -5, 6)$ in the ratio $\lambda : 1$ at point P then point P has coordinates $\left(\frac{2\lambda-1}{\lambda+1}, \frac{-5\lambda+3}{\lambda+1}, \frac{6\lambda+4}{\lambda+1} \right)$.

Point P lies on xy -plane.

$$\text{So, } \frac{6\lambda+4}{\lambda+1} = 0 \text{ or } \lambda = -\frac{2}{3}$$

Thus, xy plane divides AB externally in the ratio $2 : 3$.

11. (1) Let $(5, 5, \lambda)$ divide the segment joining $(-1, 3, 2)$ and $(-4, 2, -2)$ in the ratio $k : 1$.

$$\therefore \left(\frac{-4k-1}{k+1}, \frac{2k+3}{k+1}, \frac{-2k+2}{k+1} \right) \equiv (5, 5, \lambda)$$

$$\text{So, } -4k-1 = 5k+5 \text{ and } 2k+3 = 5k+5$$

$$\therefore k = -2/3$$

So, comparing z -coordinate with $k = -2/3$, we get

$$4/3 + 2 = \lambda/3$$

$$\therefore \lambda = 10$$

12. (3) Distance from origin $= \sqrt{1+4+9} = \sqrt{14}$

$$\text{and distance from } y\text{-axis} = \sqrt{1+9} = \sqrt{10}$$

13. (4) Distance from x -axis $= \sqrt{y^2 + z^2} = \sqrt{4+9} = \sqrt{13}$

$$\text{Distance from } y\text{-axis} = \sqrt{1+9} = \sqrt{10}$$

$$\text{Distance from } z\text{-axis} = \sqrt{1+4} = \sqrt{5}$$

$$14. (3) \quad 0 = \frac{a-2+4}{3} \Rightarrow a = -2$$

$$0 = \frac{1+b+7}{3} \Rightarrow b = -8$$

and $0 = \frac{3-5+c}{3} \Rightarrow c = 2$

15. (2) For xy -plane, $z = 0$.

$$\text{So, } \frac{6\lambda+1}{\lambda+1} = 0 \Rightarrow \lambda = -\frac{1}{6}$$

$$\therefore x = \frac{-5+18}{5} = \frac{13}{5}, y = \frac{-1+24}{5} = \frac{23}{5}$$

16. (1) Let point be (x, y, z) . Then

$$\begin{aligned} x^2 + y^2 + z^2 &= (x-a)^2 + y^2 + z^2 \\ &= x^2 + (y-b)^2 + z^2 \\ &= x^2 + y^2 + (z-c)^2 \end{aligned}$$

$$\text{Therefore } x = \frac{a}{2}, y = \frac{b}{2} \text{ and } z = \frac{c}{2}$$

17. (4) Required direction cosines are

$$\frac{3}{\sqrt{3^2+12^2+4^2}}, \frac{12}{\sqrt{3^2+12^2+4^2}}, \frac{4}{\sqrt{3^2+12^2+4^2}}$$

i.e., $\frac{3}{13}, \frac{12}{13}, \frac{4}{13}$.

18. (4) Given $\alpha = 45^\circ, \beta = 60^\circ$.

$$\text{Now, } \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\therefore \cos^2 \gamma = 1 - \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

$$\Rightarrow \gamma = 60^\circ \text{ or } 120^\circ.$$

19. (2) Here $AB = BC = CD = DA = 7$.

$$\text{Also, } AC = \sqrt{16+25+81} = \sqrt{122}$$

$$\text{and } BD = \sqrt{64+1+9} = \sqrt{74}$$

Thus, points are vertices of rhombus.

20. (4) Centre of the sphere is $C(2, 3, -4)$ and radius is $r = \sqrt{4+9+16-(-7)} = 6$.

$$\text{For point } P(1, 2, 0), CP = \sqrt{1+1+16} = \sqrt{18} < 6$$

So, point $P(1, 2, 0)$ lies inside the sphere.

For other points, we have $CP > 6$.

21. (2) For sphere $x^2 + y^2 + z^2 = 36$, centre is $C_1(0, 0, 0)$ and radius is $r_1 = 6$.

For sphere $x^2 + y^2 + z^2 - 20x + 84 = 0$, centre is $C_2(10, 0, 0)$ and radius is $r_2 = 4$.

$$C_1C_2 = 10 \text{ and } r_1 + r_2 = 6 + 4 = 10$$

Thus, two spheres touch externally at point $(6, 0, 0)$.

22. (2) Given sphere is $x^2 + y^2 + z^2 - 2x - 4y + 6z - 4 = 0$

Since it cuts the xy -plane where $z = 0$, equation of circular section is $x^2 + y^2 - 2x - 4y - 4 = 0$.

$$\text{The radius of the circle is } \sqrt{1+4-(-4)} = \sqrt{9} = 3.$$

23. (1) We have plane $6x + 3y - 2z = 18$.

Coordinates of A, B, C are, respectively, $(3, 0, 0)$, $(0, 6, 0)$ and $(0, 0, -9)$.

Thus, centroid of $\triangle ABC$ is $\left(\frac{3}{3}, \frac{6}{3}, \frac{-9}{3}\right)$ or $(1, 2, -3)$.